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Architectural Reconfiguration and Governance Capacity in Digitally Fragmented Multi-Tenant Buildings

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Abstract

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Muhammad Umar Farooq Bajwa

Buildings are increasingly equipped with sensors, automation platforms, energy-management systems and cloud services. Yet the presence of digital technologies does not necessarily make a building easier to govern. In multi-tenant buildings, digital systems are often introduced incrementally, by different actors and for different purposes, and the result can be a fragmented infrastructure in which information exists but is distributed across platforms, vendors, interfaces and organizational roles. This thesis examines how such fragmentation shapes governance capability in a multi-tenant commercial building, using Majorstuhuset in Oslo as a qualitative single case.

The study asks: How does digital fragmentation of building infrastructure shape governance capability in multi-tenant commercial buildings? Two sub-questions examine how fragmentation and infrastructural integration affect the distribution of information, knowledge and coordination responsibilities, and under what conditions integrated digital infrastructures can transform data visibility into operational actionability. The empirical material consists of four semi-structured interviews with actors occupying strategic, operational, technical-integration and engineering positions, together with technical observation, document analysis, system mapping and a limited prototype-oriented investigation of data access.

The analysis connects digital infrastructure theory, platform governance, data governance and organizational coordination, and develops a conceptual chain from infrastructure configuration to information accessibility, interpretability, operational actionability and governance capability. The findings suggest that fragmentation does not simply reduce the amount of available information; rather, it distributes information and knowledge among systems and actors, increasing the coordination work required to determine what information means, who is responsible and what action should follow. Integration can reduce these burdens, but integration alone does not create value. Data becomes operationally useful when it is accessible to the relevant actor, sufficiently contextualized and interpretable, available at the right time, connected to a clear responsibility and accompanied by sufficient ability to intervene.

The thesis therefore argues for a governance-oriented approach to building digitalization. Rather than treating integration as an end in itself, building owners should identify decisions and actions that are constrained by fragmented information and design the smallest coherent information architecture capable of supporting them. The resulting contribution is a conceptual and practical framework for understanding digital infrastructure as part of the governance capacity of existing multi-tenant buildings.

Keywords: digital infrastructure, governance capability, digital fragmentation, integration, operational actionability, multi-tenant commercial buildings

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Statement about AI

This thesis was written independently by the author, and no content was generated by AI tools. However, AI tools were used to improve English grammar and spelling. A more detailed description of the use of AI tools is provided in the appendix (see Appendix F).

Popular Science Description

Modern office buildings are increasingly filled with sensors, automated systems and digital dashboards that promise to make management easier. But does more technology automatically mean easier oversight? This thesis explores that question through a real building in Oslo called Majorstuhuset.

Majorstuhuset, like many older multi-tenant commercial buildings, has grown its digital capabilities gradually. Heating, ventilation, energy monitoring and security systems have been added over the years by different suppliers, each solving a specific problem at the time. The result is a building that generates a great deal of data, but where that data does not automatically add up to a clear picture of how the building is actually doing.

This thesis studies what happens when a building is digitally rich but organizationally fragmented, and what happens when someone tries to bring some of that fragmented information together into one accessible layer. Drawing on interviews with the people who run Majorstuhuset day to day, from a board-level facilitator to an operations manager, an external integration specialist and a site engineer, the study traces how information currently moves, or fails to move, between systems, vendors and people.

The findings show that having sensors and platforms is not the same as being able to govern a building well. Information is often scattered across many different vendors' systems, and the knowledge needed to interpret that information is scattered too. When something goes wrong, solving the problem often depends less on data than on knowing which of several vendors to call, and trusting them to give a straight answer.

The thesis also examines a practical attempt to bring some of this fragmented information together, using a small building-side computer, an "edge device", to collect and display data from several existing systems in one place without replacing them. This experiment shows that connecting systems technically is only the first step. Getting genuinely useful oversight from a building also requires the information to be timely, translated into plain language, and connected to someone who has both the knowledge and the authority to act on it.

The study concludes that building owners looking to digitalize should not simply chase more data or more integration for its own sake. Instead, they should start with the specific decisions and problems that fragmented information makes difficult, and build only the amount of

connected infrastructure needed to solve them. In an industry often focused on adding new sensors and platforms, the thesis suggests that governing a building well is, in the end, less about how much data a building has, and more about whether the right people can get the right information, at the right time, to act on it.

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1. Introduction

This chapter introduces the problem examined in the thesis. It explains why the growing availability of building data does not automatically improve the ability to govern a building, describes how fragmented digital infrastructure turns this into a governance question, and introduces Majorstuhuset as the empirical setting. The chapter ends with the research problem, the research questions, and the scope and delimitations of the study.

1.1 When a building has data but cannot use it

A building can have more information about itself than ever before and still be difficult to understand. Temperature sensors may continuously produce readings. Ventilation systems may expose operating parameters. Energy meters may generate historical consumption data. Building-management platforms may display alarms and trends, while cloud-based services can provide additional information about indoor environmental conditions, occupancy, and equipment performance. From a technical perspective, the building can therefore appear highly digitalized. Yet the existence of multiple digital systems does not necessarily mean that the information generated by these systems can be easily accessed, interpreted, coordinated, or translated into operational action.

The increasing digitalization of buildings has created new possibilities for monitoring, optimization, automation, and decision-making. Building management systems increasingly combine information and communication technologies, sensors, data processing, and analytical methods to support building operation (Taboada-Orozco et al., 2024). The International Energy Agency's work on data-driven smart buildings similarly emphasizes the growing importance of data, digital infrastructure, and interoperability for improving the operation and performance of buildings (White et al., 2025). These developments indicate that data availability is becoming an increasingly important component of building management.

However, the practical question is different: can the people responsible for the building obtain the information they need, understand what it means, coordinate with the relevant actors, and act before a problem becomes costly? Digitalization becomes operationally meaningful only when information can participate in organizational processes. A sensor reading that exists but cannot be accessed by the relevant actor has limited practical value. Similarly, information that can be accessed but cannot be interpreted in its operational context may not support an effective decision. Data visibility should therefore not automatically be equated with organizational capability.

This distinction is particularly important when considering digital infrastructure. Star and Ruhleder (1996) argue that infrastructure cannot be understood simply as a technical object; it becomes infrastructural in relation to the practices and communities that use it. Infrastructure is embedded within existing arrangements and develops through relationships between technical systems and organizational activities. Tilson et al. (2010) similarly conceptualize digital infrastructures as relational arrangements that evolve over time and shape how information systems and organizational activities interact. These perspectives are relevant to existing buildings because their digital environments are rarely created as a single, coherent system.

The central issue examined in this thesis is therefore not whether buildings have become digital. Rather, it is how the configuration of digital infrastructure influences the ability of building actors to make use of the information that digital systems produce. This shifts the focus from the simple presence of technology toward the relationship between infrastructure, information, coordination, and action.

1.2 Digitalization has increased information—but not necessarily governance capability

Digitalization has created significant opportunities for building management. Connected sensors can provide continuous information about indoor environmental conditions and equipment performance. Building-management systems can facilitate monitoring and control. Historical data can support performance analysis, while data-driven techniques can contribute to energy optimization, predictive maintenance, and automation. A recent systematic literature review by Taboada-Orozco et al. (2024) demonstrates the breadth of research surrounding data-driven building management systems and identifies the increasing role of information and communication technologies in the development of smart buildings.

The increasing amount of available information, however, introduces an important organizational question: does greater information availability necessarily increase the ability to govern building operations?

The answer cannot be assumed to be yes. Information required for a single operational decision may be distributed across several systems. The person who notices an abnormal condition may not be the person who can diagnose it. The person who controls a particular data source may not be the person who needs the information. A building operator may receive an alarm but lack the technical knowledge required to determine its significance. Conversely, an external vendor may possess the knowledge required for diagnosis but only become involved after an operational problem has already developed.

This means that the value of building data depends partly on the organizational context in which it is used. Khatri and Brown (2010) describe data governance as involving organizational decisions concerning data accountability, access, quality, and decision rights. Their work demonstrates that treating data as an organizational asset requires more than collecting and storing it. Appropriate governance arrangements are required to determine how information is managed and used. Although their framework was developed in the broader information-systems context, the principle is relevant to digitally fragmented buildings: data does not create operational value independently of the actors, responsibilities, and processes surrounding it.

The distinction can be expressed through three related concepts: visibility, interpretability, and actionability. Visibility concerns whether relevant information can be accessed or observed. Interpretability concerns whether the information can be understood within the appropriate technical and operational context. Actionability concerns whether information reaches an actor who has the responsibility, authority, and practical means to act on it, in time and in a form relevant to the decision at hand. These concepts are positioned in relation to existing literature in Section 3.6.

These dimensions are not automatically achieved through greater digitalization. A building can have extensive sensor coverage while still requiring manual inspection, specialist interpretation, or communication between several organizations before an operational decision can be made. Consequently, the practical objective of digitalization should not necessarily be to maximize the amount of available data. It should be to establish information flows that support the decisions and actions that matter for building operation.

The empirical setting of Majorstuhuset illustrates this tension. Interview material describes operational work involving manual inspection, telephone or SMS communication, experience-based identification of relevant suppliers, and dependence on external technical expertise. At the same time, the building contains digital systems and sensor sources capable of producing substantially more information. The resulting problem is therefore not simply a lack of information. It concerns whether the existing infrastructure connects information with the people, knowledge, responsibilities, and actions required to use it effectively.

1.3 The problem of fragmented building infrastructure

Existing buildings provide a particularly relevant context for studying this problem because their digital infrastructures normally develop incrementally. Systems may be introduced during construction, renovation, equipment replacement, tenant adaptation, energy projects, or changes

in service providers. Each intervention may address a specific operational requirement without considering the architecture of the building as a whole. Over time, the building can therefore accumulate technologies from different generations, suppliers, and organizational arrangements.

This characteristic corresponds with the concept of the installed base in infrastructure research. Star and Ruhleder (1996) emphasize that infrastructures develop in relation to existing systems and practices rather than being designed independently of them. Hanseth and Lyytinen (2010) similarly describe information infrastructures as heterogeneous and evolving socio-technical systems whose development is influenced by path dependency, network effects, and distributed control. Consequently, attempts to introduce new technologies into an existing infrastructure must often accommodate technologies and organizational arrangements that were not originally designed to work together.

In a building environment, this development can result in multiple parallel information environments. Some systems may communicate through relatively open standards, while others may rely on proprietary interfaces or vendor-specific platforms. Some information may be available locally, while other information is held in cloud services. Some systems may provide both monitoring and control capabilities, whereas others may only expose limited data. Access may also depend on specialist knowledge held by external suppliers.

Digital fragmentation can therefore be understood through several connected dimensions.

First, technical fragmentation concerns differences in communication protocols, interfaces, network structures, hardware, and software architectures. Systems may technically perform similar functions while relying on different methods of communication.

Second, informational fragmentation concerns the distribution and representation of data across different systems. Information may be stored in separate databases or platforms, use different naming conventions, or be structured according to different semantic assumptions. Consequently, two systems may technically exchange information while still requiring additional interpretation before that information becomes meaningful across contexts.

Third, organizational fragmentation concerns the distribution of responsibility among building owners, facility managers, operators, service providers, specialist vendors, and tenants. Different actors may control different systems or possess different forms of technical

knowledge. The distribution of technical infrastructure can therefore correspond with a distribution of organizational knowledge and responsibility.

Fourth, procedural fragmentation concerns the routines through which problems are detected, communicated, diagnosed, and resolved. Different actors may use different procedures for responding to information, meaning that technical connectivity alone does not guarantee coordinated action.

These dimensions can reinforce one another. A technically isolated system may require a specialist vendor. Dependence on that vendor can preserve organizational separation. Organizational separation may make documentation more difficult to maintain, particularly when systems are repeatedly modified or personnel change. Poor documentation can then increase the difficulty of future integration. Fragmentation is therefore better understood as a socio-technical condition rather than simply as an interoperability defect.

This perspective also changes the way integration should be evaluated. If fragmentation is produced through the interaction of technical and organizational arrangements, then integration cannot be reduced to connecting devices or transferring data between interfaces. Effective integration may require common representations, documentation, appropriate access rights, organizational responsibility, and the ability to interpret information across system boundaries.

1.4 From technical fragmentation to governance

Governance is used in this thesis in a practical organizational sense. Governance capability refers to the ability of relevant building actors to access, interpret, coordinate around, decide upon, and act on information necessary for managing the building. This definition deliberately avoids treating governance as synonymous with formal hierarchy or ownership. A building may have a clearly defined owner and still have weak governance capability if essential information is inaccessible, difficult to interpret, distributed among several actors, or dependent on external specialists.

The concept is particularly relevant to digital infrastructure because infrastructure influences the pathways through which information can move. Interfaces, protocols, data formats, access rights, and platforms can make particular forms of coordination easier while making others more difficult. Tilson et al. (2010) argue that digital infrastructures should be understood as relational constructs that shape organizational activities, while Hanseth and Lyytinen (2010) emphasize that infrastructures are heterogeneous, evolving, and subject to distributed forms of control. From this perspective, infrastructure is not merely a technical foundation beneath

organizational activity. Its configuration can influence how organizational activity is performed.

Star and Ruhleder's (1996) infrastructure perspective is also useful here because it highlights the relationship between infrastructure and practice. An infrastructure becomes visible when it breaks down or when users encounter difficulties in accessing or using it. In a building, this can occur when an operator cannot determine which system contains the relevant information, when a vendor must be contacted to interpret a technical condition, or when information exists but cannot be connected to an appropriate operational response.

At the same time, integration should not be treated as an unconditional solution. Greater connectivity can introduce new dependencies, cybersecurity concerns, complexity, and responsibilities. The objective is therefore not to eliminate every boundary between systems. Rather, it is to determine which information flows are necessary for effective operational decisions and design appropriate connections between them.

The central analytical proposition of this thesis is that digital infrastructure affects governance indirectly through information and action. Fragmentation can distribute information and knowledge across systems and actors, increasing coordination requirements. Integration can reduce some of these burdens, but only when the resulting information is meaningful to the actors who must use it.

The thesis therefore develops the following conceptual sequence: digital infrastructure configuration shapes information accessibility, which shapes interpretability, which shapes operational actionability, which in turn shapes governance capability. This sequence is developed formally as a conceptual model in Figure 1 (Section 3.6).

This sequence is important because it distinguishes integration from value. Integration is not valuable simply because more systems communicate. Access is not valuable simply because more information becomes visible. The value of integration depends on whether it improves the ability of relevant actors to understand a situation and take appropriate action.

1.5 Majorstuhuset as the empirical setting

Majorstuhuset is a multi-tenant commercial office building in Oslo with digital systems that have developed incrementally over time. Like many existing buildings, it contains multiple technical systems from different vendors while tenant networks remain separate from building-

owned infrastructure. This fragmentation creates a real-world setting where questions of integration, access, responsibility, and coordination can be examined.

The case is particularly relevant because different building actors require different forms of information. A facilitator may need investment-level summaries, an operations manager may need immediate alerts about failures, and a specialist may need detailed technical parameters. This demonstrates why "access to data" cannot be a one-size-fits-all requirement—what matters is whether the right information is available in the appropriate form for each actor's task.

The practical investigation included examination of the Lenovo ThinkEdge SE10 as a building-side edge-computing concept, Node-RED, Airthings cloud integration, and existing building systems. It also examined integration boundaries with protocols such as BACnet and Modbus through an external specialist. The technical investigation provides evidence about the possibilities and limitations of integration in an existing building context.

Importantly, the thesis distinguishes between what was directly implemented or observed and what remained proposed. Some elements (such as MQTT and ESP32 nodes) represent architectural exploration rather than completed implementation, and this distinction is maintained throughout to avoid overstating outcomes.

1.6 Research problem and gap

Existing research provides substantial knowledge about smart-building technologies, building-management systems, interoperability, data-driven operation, and digital infrastructure. Smart-building research increasingly examines how sensors, data analytics, automation, and connected systems can support building performance and operational intelligence (Taboada-Orozco et al., 2024). The literature on digital infrastructure, meanwhile, provides strong theoretical explanations of how heterogeneous and evolving infrastructures become embedded within organizational practices (Star & Ruhleder, 1996; Tilson et al., 2010; Hanseth & Lyytinen, 2010).

These research areas nevertheless emphasize somewhat different problems. Smart-building research frequently focuses on technological capabilities, energy performance, monitoring, optimization, and automation. Digital infrastructure research focuses more broadly on the evolution, organization, and governance of information infrastructures. Data-governance research emphasizes the organizational responsibilities and decision structures surrounding information (Khatri & Brown, 2010).

The gap addressed by this thesis lies at the intersection of these perspectives. The study focuses on the governance implications of fragmented digital infrastructure in an existing multi-tenant building. The question is not only whether systems can technically communicate, but what happens to information, knowledge, coordination, and responsibility when systems remain separated, and what changes when information is reorganized into a more coherent architecture.

This framing treats digital integration as more than a technical upgrade. It treats infrastructural reconfiguration as a potential organizational and governance intervention. The thesis therefore examines the conditions under which greater data visibility can become operational actionability and, ultimately, contribute to governance capability.

The contribution is consequently not a claim that all building systems should be integrated into a single platform. Instead, the thesis investigates a more specific problem: under what infrastructural and organizational conditions can integration improve the ability of building actors to coordinate information and act upon it?

1.7 Research questions

The main research question is:

How does digital fragmentation of building infrastructure shape governance capability in multi-tenant commercial buildings?

The study addresses two sub-questions:

RQ1: How do fragmentation and infrastructural integration affect the distribution of information, knowledge, and coordination responsibilities among building actors?

RQ2: Under what conditions can integrated digital infrastructures transform data visibility into operational actionability?

Together, these questions move the analysis from the existence of digital technologies toward their organizational consequences. The main research question establishes the relationship between digital fragmentation and governance capability. RQ1 examines how fragmentation and integration influence the distribution of information, knowledge, and coordination responsibilities. RQ2 focuses on the conditions under which integrated infrastructure can convert information visibility into practical operational action.

Methodologically, the research questions are addressed through a qualitative single-case study of Majorstuhuset (Chapter 4). The main research question and RQ1 are examined through four

semi-structured interviews with actors in strategic, operational, technical-integration and engineering positions, supported by micro-ethnographic field observation, system mapping and documentary material. Together, these sources show how information, knowledge and coordination responsibilities are distributed among systems and actors. RQ2 is examined by combining the interview accounts with a design-oriented technical investigation in which a building-side integration layer was used to access and reorganize selected building data. This made it possible to observe what changes when information becomes more visible, and what is still required for it to become actionable. The material is analyzed through reflexive thematic analysis informed by the framework developed in Chapter 3.

1.8 Scope and delimitations

The thesis is a qualitative single-case study of Majorstuhuset. It does not attempt to quantify the building's energy savings, conduct a complete cybersecurity audit, or demonstrate that one particular technical architecture is universally optimal. Nor does it claim that every subsystem in the building was integrated.

The technical investigation and prototype work are treated as empirical and design-oriented evidence concerning possibilities, constraints, and integration boundaries rather than as proof of a completed building-wide digital transformation. Similarly, information obtained from interviews, project documentation, and direct observation is distinguished according to its source during the empirical analysis.

The study also does not assume that all information should be visible to all actors. Appropriate information access must remain compatible with cybersecurity, role responsibilities, organizational boundaries, and legitimate operational requirements. The objective is therefore not maximum openness, but appropriate accessibility combined with coherent coordination.

The study further recognizes that integration can create new dependencies as well as reduce existing ones. Connecting previously separate systems may introduce new technical interfaces, centralized points of dependency, security requirements, and responsibilities for maintaining the integrated architecture. Integration should therefore be evaluated according to its operational purpose rather than treated as an inherently positive outcome.

The objective of the thesis is consequently not to propose a universal smart-building architecture. Instead, it is to develop an empirically grounded understanding of how digital fragmentation shapes governance capability and under what conditions infrastructural integration can improve the transformation of information into action.

The remainder of the thesis develops this argument through a review of literature on digitalization, digital infrastructure, interoperability, data governance, and multi-actor building operations; a theoretical and conceptual framework linking fragmentation with governance capability; a qualitative investigation of Majorstuhuset; and an analysis of how infrastructural configuration shapes information accessibility, interpretability, actionability, and coordination.

2. Literature Review

This chapter reviews the literature on which the study builds. It moves from the digitalization of the built environment and the character of digital infrastructure in existing buildings, through fragmentation, interoperability and building data, to the multi-actor and vendor relationships that surround building operation and the governance questions they raise. The chapter concludes by identifying the research gap addressed by the thesis.

2.1 Digitalization of the built environment

Digitalization has changed the way buildings can be observed, operated, and managed. Activities that were traditionally handled through periodic inspections, manual measurements, local control panels, or separate specialist systems can increasingly be supported by sensors, communication networks, software platforms, analytics, and automated controls. Within the smart-building literature, these technologies are commonly associated with energy management, monitoring, indoor environmental quality, predictive maintenance, and operational efficiency (Domingues et al., 2016; Yaïci et al., 2021; Taboada-Orozco et al., 2024). Building automation has consequently developed from relatively isolated control functions toward environments in which information can be collected from several technical domains and used for more data-driven forms of operation.

The literature nevertheless presents digitalization as a broad and heterogeneous development Taboada-Orozco et al. (2024), in a systematic review of data-driven building management systems, describe a field increasingly combining information and communication technologies, sensing, data processing, and artificial intelligence. Yaïci et al. (2021) similarly review IoT infrastructures for building energy systems as combinations of sensors, actuators, communication technologies, monitoring, and control. The technical opportunity is therefore not based on one device or software package, but on the ability of several components to work together.

Recent work on smart-building systems also emphasizes that digital technologies increasingly cross traditional system boundaries. A sensor may belong physically to one subsystem but its data may be consumed by another application, a cloud service, an analytics model, or a human operator. As digitalization expands, the building becomes a network of technical and organizational relationships rather than a collection of independent automation systems. This increases the importance of architecture: how information moves, where it is processed, and who can use it can become as important as the sensing technology itself.

The literature also suggests that digitalization needs to account for users and context. Building information does not have one universal purpose. A specialist may require detailed equipment values, while an operator may need a trend or an alarm, and a decision-maker may need only a high-level indication that intervention is justified. Digitalization can therefore create new possibilities while simultaneously creating a requirement to organize information around actual practices and decisions. This provides the first bridge from the technical promise of smart buildings toward the organizational questions examined in this thesis.

2.2 Digital infrastructure in existing buildings

The concept of digital infrastructure provides a broader way of understanding digitalization. Rather than treating sensors, control systems, databases, and platforms as isolated technologies, infrastructure research focuses on relationships among technical artefacts, standards, users, organizations, and established practices. Star and Ruhleder (1996) describe infrastructure as relational and embedded: it becomes infrastructural in relation to the practices and communities that depend upon it. Infrastructure is therefore not simply a technical object but an arrangement experienced through use, access, interdependence, and breakdown.

This perspective is particularly relevant to existing buildings because their digital systems are rarely designed simultaneously as one coherent architecture. Heating, ventilation, cooling, energy monitoring, access control, indoor-environment sensing, and other systems are often introduced at different times, for different operational requirements, and by different suppliers. The building therefore contains an installed base rather than a single digital system. Existing components and practices become part of the conditions that future changes must accommodate (Star & Ruhleder, 1996; Hanseth & Lyytinen, 2010).

Tilson et al. (2010) conceptualize digital infrastructures as shared, heterogeneous, and evolving arrangements that cross organizational boundaries. Their emphasis on infrastructures as distinct from individual applications is particularly important because infrastructures are built incrementally and must simultaneously support stability and future change. Hanseth and Lyytinen (2010) similarly explain how information infrastructures develop through path dependency and dynamic complexity. Earlier architectural choices influence what can be added later, and attempts to redesign an infrastructure must account for the dependencies accumulated over time.

This installed-base perspective changes the meaning of modernization. In a new building, designers may be able to specify common protocols, data models, networks, and control layers

before equipment is installed. In an existing building, digitalization is more likely to involve adaptation, translation, bridging, and gradual replacement. A gateway may connect a legacy controller to a new application. An API may expose information from an existing platform without replacing it. A local processing layer may collect information from multiple sources while leaving the underlying systems intact. These are examples of infrastructural reconfiguration rather than clean-slate design.

The question of how digital components are layered is also relevant to this reconfiguration. Yoo et al. (2010) describe a layered modular architecture in which digital technologies create loosely coupled layers of devices, networks, services, and content. Their argument is important for this study because it shows how digital innovation can be organized by creating new layers above existing components rather than replacing all underlying assets. In a building context, this distinction allows an integration layer to coexist with existing devices and control systems while enabling new services and information uses.

The organizational dimension of infrastructure is equally important. Orlikowski (1992) argues against deterministic views of technology by emphasizing the interaction between technological properties and human action. Orlikowski and Scott (2008) develop a sociomaterial perspective in which technology, work, and organization are understood as mutually implicated. From this perspective, adding an integration layer does not simply change the technical architecture. It can also change who sees information, who performs coordination work, and what kinds of organizational practices become possible.

2.3 Fragmentation and interoperability

If digital infrastructure evolves as an installed base, fragmentation becomes a natural possibility. Fragmentation occurs when relevant technical systems, information sources, platforms, and organizational responsibilities remain separated rather than forming a coherent arrangement. In building environments, this may be visible through multiple equipment generations, vendor-specific applications, different communication protocols, disconnected databases, or information that is technically available but difficult to combine. Domingues et al. (2016) identify heterogeneity and the difficulty of integrating solutions from different manufacturers as persistent problems in building automation systems.

Interoperability is therefore an important response to fragmentation, but it is not equivalent to complete integration. At the most basic level, interoperability concerns whether two systems can exchange information. However, effective interoperability may require shared

representations, common models, and knowledge of how information should be interpreted after it crosses a system boundary. Di Biccari et al. (2022), reviewing interoperability between building information modeling and building performance simulation, emphasize information-transfer processes, common modeling approaches, and exchange strategies rather than treating interoperability as a single technical connection.

Research focused more specifically on smart buildings makes a similar point. Verbeke et al. (2022) argue that interoperability can act both as a driver and a barrier for smart-building technologies. They identify differences in system representations and the need for system knowledge as important constraints, while also noting that greater interoperability can introduce concerns relating to cybersecurity, malfunction, and liability. Interoperability therefore changes the conditions under which building systems operate; it does not simply remove technical barriers.

Semantic interoperability is particularly relevant to the present research. Two systems may expose values that are technically compatible while representing comparable concepts in different ways. Qiang et al. (2023) compare major building-data ontology approaches and show that they differ in scope and structure. Their review suggests that no single universal ontology resolves all building-data integration requirements and that ontology alignment and matching remain important. This provides a useful distinction between connectivity and interpretability: a value can be successfully transferred while remaining difficult to understand in relation to values from another system.

The implication is that integration has at least two dimensions. The first is connectivity: can information be retrieved, transmitted, or written between systems? The second is semantic and operational integration: can the receiving system or actor understand what the information represents, relate it to other information, and use it to support a decision? The second dimension is especially important for governance because the purpose of exchanging building data is ultimately to support a process of interpretation, coordination, and action.

The layered architecture discussed by Yoo et al. (2010) helps explain why these dimensions can be separated. Devices may remain unchanged while a new network or service layer is added. Similarly, a building can create a new information service above legacy equipment without requiring every underlying device to communicate directly with every other device. This provides flexibility, but it does not eliminate the need for common meanings and appropriate interfaces between layers.

2.4 Building data, information quality, and accessibility

As buildings become increasingly instrumented, data quality and usability become more important than data quantity alone. Data can differ in timeliness, context, granularity, and accessibility. A technically correct measurement is not useful if the user does not know where it originated, what it represents, or how it should influence a decision.

The literature on building information increasingly emphasizes structured representations. Qiang et al. (2023) show that ontologies support data exchange and integration, while demonstrating substantial differences among existing approaches. A common information environment needs more than a common transport protocol—the relationships among sensors, equipment, zones, and measurements also need consistent representation.

Data governance provides an organizational complement to this technical perspective. Khatri and Brown (2010) describe data governance in terms of decision domains and accountability around data. This suggests that access should not be viewed merely as a technical permission: someone must have the ability and responsibility to maintain information, determine access, and ensure appropriate use.

In a building, this raises practical questions: Who can retrieve the data? Who knows what it means? Who is responsible if information is missing? What happens when the expert who understands the system leaves? These questions connect information accessibility with organizational memory.

Data value should therefore be treated carefully. A dataset that is technically rich but difficult to interpret may have lower practical value than smaller data that is reliable, timely, contextualized, and directly linked to an operational decision. This distinction supports the thesis's focus on the movement from data visibility toward operational actionability.

2.5 Multi-actor building operations

A multi-tenant building is not governed through technology alone. Its operation involves owners, facilitators, operations personnel, caretakers, tenants, engineers, consultants, equipment suppliers, service companies, software providers, and integration specialists. These actors possess different responsibilities, knowledge, resources, and information requirements. The resulting environment is therefore one of interdependence rather than simple hierarchy.

No single actor necessarily possesses all the capabilities required to operate and transform a complex building. The building owner may have strategic authority but depend on vendors for

specialist knowledge. An operations manager may understand tenant needs and recurring problems but depend on technical contractors for diagnosis. An integration specialist may understand protocols and interfaces but not have authority over building investment decisions. Engineers may connect technical knowledge across disciplines while still relying on suppliers for detailed subsystem knowledge. The theoretical implications of this interdependence are developed through the ecosystem perspective in Section 3.2.

2.6 Vendor dependency and organizational knowledge

Vendor dependency is a recurring concern in digital platforms because proprietary interfaces and standards can influence switching costs and control. In a building environment, similar mechanisms emerge when a supplier controls a data gateway, programming interface, or knowledge base.

However, the ecosystem perspective suggests that some dependency may be functional. Jacobides et al. (2018) show that ecosystems involve interdependence among actors—such dependency may provide access to capabilities expensive to internalize, but it becomes problematic when one actor constrains the rest of the system.

Building infrastructures intensify these issues because physical components have long lifetimes. Replacing a deeply installed system requires physical work, commissioning, training, and contractual changes. Switching suppliers therefore becomes more consequential than replacing software. Moreover, dependency can be knowledge-based: a system may be technically accessible through an open protocol, but only a particular contractor understands its configuration.

Documentation is therefore not merely administrative. It preserves knowledge across organizational boundaries and time. Good documentation allows a new supplier to understand a configuration, helps internal staff challenge proposals, and reduces reconstruction when systems change. In infrastructural terms, documentation contributes to legibility of the installed base.

The empirical case requires a balanced interpretation of vendor relationships. Specialist expertise is often valuable and sometimes necessary. The objective should not be eliminating suppliers, but ensuring the building organization retains sufficient information, access, and decision capacity to coordinate them. Governance capability may therefore depend on reducing unnecessary dependency while preserving useful interdependence.

2.7 Digital infrastructure and governance

Tilson et al. (2010) identify change and control as recurring tensions in digital infrastructures. An infrastructure must remain stable enough to support reliable operation while remaining adaptable enough to accommodate new users and applications. For an existing building, this tension can appear in the choice between tightly controlled vendor-specific platforms and more open, modular arrangements. Neither extreme is automatically optimal; the relevant governance question is how architectural choices influence flexibility, security, responsibility, and dependency.

Yoo et al. (2010) provide a useful bridge between architecture and governance through layered modularity. When devices, networks, services, and content can be loosely coupled, new services may be introduced without rebuilding the entire physical layer. This can support innovation, but it can also create new governance questions about who controls each layer, who can access information, and how changes in one layer affect the others.

Platform governance develops this argument further. Tiwana (2014) shows that architecture and governance are interdependent: interfaces, modularity, and control points influence participation and value creation. In a building, an API or open protocol may reduce a supplier's control over data access, while a centralized integration layer may create a new architectural control point. The governance issue is therefore not simply openness versus closure but where control is located and whether it supports the building's strategic objectives.

Data governance adds the question of information rights and responsibilities. Khatri and Brown (2010) emphasize decision rights and accountability around data. In a building, this can involve decisions about who can access operational data, who maintains it, who is responsible when it is incorrect, and which systems may consume it. These questions become particularly important when data crosses organizational boundaries between building owners, vendors, tenants, and technology providers.

The ecosystem literature reinforces the need for coordination across boundaries. Adner (2017) emphasizes interdependence across actors and activities, while Jacobides et al. (2018) show how modularity can coordinate independent actors without full hierarchy. This provides a useful way to think about building digitalization: the architecture can be technically modular, but governance must still align the actors whose contributions are necessary for the system to work.

Power can also be considered through this lens, but only where the empirical material supports it. Lukes (2005) distinguishes visible decision-making power from less visible forms through

which agendas and perceptions of possibility are shaped. Architectural control over interfaces, standards, or information can influence what actors are practically able to do. The more precise claim for this study is therefore not that technology determines power, but that technical arrangements can allocate practical capabilities among actors.

This leads toward the thesis's concept of governance capability. Governance is not only about formal ownership. It also depends on whether relevant actors can access information, interpret it, coordinate with others, decide, and intervene. Digital infrastructure can support or constrain these capabilities through its architecture, information flows, and dependencies.

2.8 Smart buildings, energy flexibility, and the technical promise of integration

The technical promise becomes especially relevant when information and control are connected. A sensor can detect a change, an analytics layer can interpret that change, and an actuator or control system can respond. Yet this sequence depends on the quality and interoperability of each layer. If information cannot move, if the meaning of the data is uncertain, or if a control pathway does not exist, the potential of the integrated system is reduced.

The interoperability literature identifies these limitations clearly. Verbeke et al. (2022) describe interoperability as both a driver and a barrier and note that broader integration introduces issues such as cybersecurity, liability, and system complexity. Domingues et al. (2016) similarly show that building automation remains characterized by heterogeneity and integration challenges. The literature therefore supports a cautious view of integration: it can create new capabilities, but it can also create new technical and organizational dependencies.

Semantic structure becomes increasingly important as applications become more advanced. Qiang et al. (2023) show the diversity of building ontologies, while Di Biccari et al. (2022) emphasize common modeling and information-transfer strategies. These findings suggest that advanced analytics and AI cannot simply be placed on top of arbitrary data. They require information that is structured sufficiently for relationships, contexts, and meanings to remain understandable across systems.

The same logic applies to automation. Yaïci et al. (2021) show that building-energy IoT involves both sensing and actuation. Automation therefore requires not only a signal but a defined relationship between a condition and a possible action. In governance terms, this raises a further question: who is allowed to define that relationship and who remains responsible for

oversight? Increased automation may reduce repetitive work while simultaneously increasing the need for clear rules, accountability, and human control.

The literature thus supports the technical case for integration but also reveals the limitations of purely technical approaches. A building can become more connected without becoming more governable. This is precisely the point at which the present study moves beyond asking whether systems can be integrated and instead asks how integration changes information, coordination, and action.

2.9 Research gap

The literature reviewed above contains several well-developed but only partially connected research streams. The first is smart-building and IoT research, which demonstrates the technical potential of sensors, analytics, automation, and connected building management (Domingues et al., 2016; Yaïci et al., 2021; Taboada-Orozco et al., 2024). The second is digital-infrastructure research, which explains how heterogeneous, evolving infrastructures emerge through installed bases, path dependency, and interactions between technology and organization (Star & Ruhleder, 1996; Tilson et al., 2010; Hanseth & Lyytinen, 2010). The third is digital-innovation and architectural research, which explains how layered modularity can allow new digital services to emerge from combinations of devices, networks, services, and content (Yoo et al., 2010). The fourth is ecosystem and governance research, which explains how interdependent actors, complementary activities, organizational boundaries, architecture, and data decision rights shape value creation and control (Moore, 1993; Adner, 2017; Jacobides et al., 2018; Kapoor, 2018; Khatri & Brown, 2010; Tiwana, 2014).

Each stream helps explain an important part of the problem, but the connection between them is less developed in the context of existing multi-tenant buildings. Smart-building studies demonstrate what integrated technologies may technically enable, but often place less emphasis on the organizational work required to make heterogeneous infrastructures coherent. Infrastructure research explains how installed bases evolve and why change is difficult, but is less frequently used to examine the everyday governance of building information. Ecosystem research explains interdependence among actors, but is more commonly applied to firms, industries, and market-facing innovation than to the governance of a physical building's digital infrastructure. Digital-innovation research explains modular layers, but does not by itself explain how organizations decide what information should be exposed, interpreted, or acted upon.

A second gap concerns the distinction between data availability and operational usefulness. The literature identifies the importance of interoperability, information models, data governance, and user requirements, but these issues are often discussed separately. Less attention is given to the intermediate process through which information moves from being technically visible to being interpretable and finally actionable for a specific organizational actor. This distinction becomes especially important where different actors need different degrees of information and where technical responsibility and decision authority are distributed.

A third gap concerns the governance consequences of infrastructural reconfiguration. The layered architecture described by Yoo et al. (2010) suggests that digital capability can be added through new layers rather than through complete technological replacement. Ecosystem research suggests that such reconfiguration affects a set of interdependent actors rather than one organization in isolation (Adner, 2017; Jacobides et al., 2018). In an existing building, this raises questions about who gains access to information, which actors remain necessary, where coordination work moves, and whether the new architecture reduces or merely changes dependency.

A fourth gap concerns the location of value. Ecosystem research emphasizes that value can be created through complementarities and interdependence among actors rather than by a single organization acting alone (Kapoor, 2018; Jacobides et al., 2018). In a building-information context, this suggests that data should not be treated as automatically valuable merely because it is produced. Its value depends on whether it contributes to the information and decision processes through which building actors create operational outcomes. This provides a useful basis for distinguishing data visibility from operational actionability.

This thesis addresses these gaps by bringing digital infrastructure, layered digital architecture, ecosystem thinking, and governance perspectives together in the study of a multi-tenant commercial building. The empirical focus is not on proving that integration is always beneficial. Instead, the study examines how fragmentation distributes information, knowledge, and coordination responsibilities and under what conditions a more coherent digital infrastructure can transform data visibility into operational actionability.

The literature therefore leads to the central analytical concern of this thesis: governance capability depends not only on whether information exists, but on whether the infrastructure and actor relationships allow the relevant people to access, interpret, coordinate around, and act

upon it. This provides the basis for the theoretical and conceptual framework developed in the next chapter.

3. Theoretical and Conceptual Framework

This chapter develops the theoretical and conceptual framework used to interpret the empirical material. It combines digital infrastructure theory, platform governance and the ecosystem perspective, data governance, and organizational coordination, and brings these together in the concept of governance capability. The chapter ends by introducing the distinction between visibility, interpretability and actionability that structures the later analysis.

3.1 Digital infrastructure theory

The first theoretical pillar is digital infrastructure theory. The starting point is that infrastructure should not be understood as a neutral technical substrate that simply sits beneath organizational activity. Instead, infrastructure is embedded in practices, institutional arrangements, standards, and communities of users. Star and Ruhleder (1996) show that infrastructure becomes meaningful through the work it supports and through the relationships between technical systems and the people who use them. In this sense, infrastructure is relational rather than merely physical or technical. This is particularly relevant to the present study because the research problem concerns not only whether building systems are connected, but how their configuration affects the ability of actors to access and use information.

A central concept in this perspective is the installed base. Star and Ruhleder (1996) emphasize that infrastructure is difficult to understand independently of what is already in place. New infrastructural elements are introduced into existing arrangements of hardware, software, conventions, skills, and organizational practices. Hanseth and Lyytinen (2010) develop this idea in their design theory for information infrastructures, defining infrastructures as shared, open, heterogeneous, and evolving socio-technical systems. They emphasize that infrastructure development is path dependent and nonlinear: existing choices shape future possibilities, while new additions become part of the conditions for subsequent change.

Tilson et al. (2010) similarly argue that digital infrastructures deserve analytical attention as a distinct category because they extend across organizational boundaries and evolve through continuous development and use. Infrastructures must remain sufficiently stable to support existing activity while also remaining open to future applications and users. This creates a tension between continuity and change. For a building with long-lived physical systems, the implication is that digitalization is often a process of adaptation rather than replacement.

The installed-base concept is important for how fragmentation is understood in this thesis. A fragmented building infrastructure is not necessarily the result of one poor design decision. It

can be the cumulative outcome of many locally rational decisions made at different points in time. A ventilation controller can be upgraded without redesigning the energy platform. A sensor system can be introduced without changing the legacy automation architecture. A cloud service can be adopted because it solves an immediate operational problem while creating another information boundary. Hanseth and Lyytinen's (2010) path-dependency argument helps explain why these incremental decisions can become increasingly difficult to reverse.

The infrastructure perspective also avoids technological determinism. Orlikowski (1992) argues that technology and organizational structures should not be treated as separate forces in which technology simply determines organizational outcomes. Instead, technologies are enacted through organizational practices, while those practices also influence how technologies are used and changed. Leonardi (2011) similarly shows how human and material agencies become imbricated through routines and technologies, creating infrastructure that both affords and constrains action. These perspectives are useful here because an integrated technical architecture may create new possibilities without automatically producing new organizational capabilities; actors still have to incorporate the infrastructure into their work.

For the present study, digital infrastructure theory therefore provides the foundation for treating fragmentation as an architectural and organizational condition. It directs attention to the installed base, the evolution of heterogeneous systems, and the relationship between technical arrangements and organizational practice. The key analytical implication is that fragmentation should not be examined simply as a collection of incompatible devices. It should be examined as a condition that can shape information flows, knowledge distribution, coordination, and the practical possibilities available to building actors (Star & Ruhleder, 1996; Hanseth & Lyytinen, 2010; Tilson et al., 2010).

3.2 Platform governance and ecosystem perspective

A second theoretical pillar concerns platforms, architecture, and ecosystems. This perspective is useful because the digital environment examined in this thesis includes multiple systems and actors that interact through interfaces, protocols, services, and information flows. The building is not treated as a commercial digital platform in the strict sense. Instead, platform theory is used to analyze architectural modularity, interfaces, access, dependency, and the governance of an environment in which multiple actors contribute to a broader outcome.

Yoo et al. (2010) provide an important bridge between digital architecture and organization through their concept of layered modular architecture. They argue that pervasive digitization

creates loosely coupled layers of devices, networks, services, and content. The significance of the model is not simply that digital systems have layers, but that new digital services can be developed by recombining elements across those layers. This creates opportunities to innovate without replacing every underlying component. In the context of Majorstuhuset, the idea helps conceptualize an integration layer that can obtain information from existing building systems and make it available to new applications without requiring a complete replacement of the installed base.

The layered architecture also creates governance questions. When devices, networks, services, and information are separated into modules, decisions about interfaces and access become important. A technically open interface may allow additional applications or suppliers to participate. A closed interface may increase dependence on the actor controlling it. Tiwana (2014) places this relationship between architecture and governance at the centre of platform ecosystems, arguing that architecture, governance, and strategy must evolve together. The relevant issue for this study is therefore not simply whether an interface exists, but how interface design affects participation, control, flexibility, and dependence.

The ecosystem perspective extends this analysis beyond one platform or one organization. Moore (1993) introduced the business-ecosystem concept to describe environments in which organizations co-evolve and depend on suppliers, partners, customers, and other actors rather than operating in isolation. Adner (2017) later developed an ecosystem-as-structure view centered on interdependence among actors and activities. Jacobides et al. (2018) distinguish ecosystems from markets, supply chains, and hierarchies and emphasize modularity, complementarities, multilateral dependencies, and shared rules as mechanisms through which independent organizations can coordinate.

These ideas are applicable to Majorstuhuset as an analytical lens rather than as a claim that the building itself constitutes a business ecosystem in the strict strategic-management sense. The building's operation depends on complementary contributions from owners, operational staff, engineers, system vendors, integration specialists, and other service providers. The relevant point is that no single actor necessarily possesses all the capabilities required for digital building management. The building's governance is therefore influenced by interdependence among actors who contribute different forms of knowledge, technology, and authority (Adner, 2017; Jacobides et al., 2018).

Kapoor (2018) emphasizes that ecosystem thinking broadens the locus of value creation by focusing on complementarities and interdependence. Gomes et al. (2021) similarly conceptualize ecosystems as forms of meta-organization in which distributed actors contribute to outcomes that cannot be explained by one organization alone. This is relevant to the present study because the operational value of building data may depend on a chain of complementary activities. A sensor provider may produce the data, an integration specialist may expose it, an engineer may interpret it, an operations manager may coordinate a response, and a decision-maker may authorize an investment. The resulting value is therefore distributed across actors rather than residing solely in the technical data source.

The ecosystem perspective also helps qualify the idea of vendor dependency. Interdependence is not inherently negative. It can provide access to specialized capabilities that would be costly to develop internally (Jacobides et al., 2018; Kapoor, 2018). The governance question is whether interdependence remains manageable or whether one actor gains disproportionate control over critical information or infrastructure. This distinction is important for the thesis because the empirical material does not suggest that vendors should disappear. Rather, it raises questions about how the building organization can retain sufficient knowledge, access, and decision rights while still benefiting from specialist expertise.

Taken together, layered architecture, platform governance, and ecosystem theory provide three connected analytical questions. First, how can new digital capabilities be added to an existing technical base? Second, how do interfaces and architectural choices shape access and dependency? Third, how do interdependent actors contribute to and govern the resulting system? These questions help connect the technical structure of integration with the organizational relationships through which its value is realized (Yoo et al., 2010; Tiwana, 2014; Adner, 2017; Jacobides et al., 2018).

3.3 Data governance

The third theoretical component concerns data governance. Khatri and Brown (2010) describe data governance as the exercise of authority and control over the management of data assets and identify decision domains and accountability as important elements of this governance. Their starting point is that organizations increasingly view data as an asset but require explicit structures for deciding how it should be managed and used. This perspective is useful for buildings because data may be generated by one technical system, stored by another organization, and used by yet another actor for operational decisions.

Data governance in this thesis is therefore broader than privacy or access control. It includes the organizational arrangements through which data becomes an effective resource: who can access it, who maintains it, who is responsible for its quality, which systems can consume it, and who can act upon information derived from it. Khatri and Brown (2010) demonstrate that data governance concerns decision rights and accountability, which provides a useful way to examine the relationship between information ownership and operational usability.

This distinction between ownership and usability is important. A building may formally own data while lacking a convenient mechanism for retrieving it. Conversely, a supplier may provide access to a dashboard without transferring the underlying data, system knowledge, or configuration capability. The governance question is therefore not simply who owns the data, but where practical control over its use resides. Tiwana (2014) similarly demonstrates that architecture and governance are intertwined in platform environments because control over interfaces and modular components affects participation and dependency.

Data governance also provides a basis for examining different information requirements among actors. A senior decision-maker may need a concise signal that a system requires attention, while an operations manager may need historical data and trends, and a technical supplier may require detailed controller-level information. Treating data governance as a question of appropriate decision rights and accountability allows these differences to be recognized without assuming that equal access for everyone is always the objective (Khatri & Brown, 2010).

The thesis therefore uses data governance to examine how information access intersects with organizational responsibility. A technically accessible dataset has greater governance relevance when the actor receiving it understands its significance, has a legitimate responsibility to use it, and can connect it to a decision or action. Conversely, information can remain operationally weak when access exists but responsibility, meaning, or authority is unclear.

This perspective also creates space for security as a governance condition. More open information flows can support interoperability and innovation, but they increase the importance of determining who should have access and what actions different actors are authorized to perform. Governance-enhancing integration therefore requires a balance between accessibility, accountability, and protection rather than unrestricted openness (Khatri & Brown, 2010; Tiwana, 2014).

3.4 Organizational coordination and dependency

The fourth theoretical component concerns coordination. Organizations coordinate because activities and knowledge are interdependent. In the present case, the relevant interdependencies are both technical and organizational: a building problem may require information from one system, interpretation by a specialist, communication with another actor, and authorization from a decision-maker. Fragmentation can increase the number of steps required to move from detection to action because information and responsibility are distributed across different parts of the system.

Barley (1986) provides an important foundation for examining this relationship. His study of CT scanners showed that the same technology could become associated with different organizational structures depending on how roles and interactions developed around it. The implication is that technology can occasion organizational restructuring without mechanically determining one organizational outcome. This is relevant to digital building infrastructure because a new integration layer can change who receives information, who is contacted, who performs diagnostic work, and how responsibilities are allocated.

Leonardi (2011) offers a complementary perspective by showing how technologies and routines become mutually configured over time. People perceive technologies as constraints or affordances in relation to their existing practices, while their responses can also lead to changes in technology and routines. In a fragmented building, informal routines such as knowing which vendor to contact can function as part of the operational infrastructure. Digital integration may therefore change coordination by making some of these routines less necessary, but it can also create new routines around dashboards, alerts, data interpretation, and maintenance of the integration layer.

From this perspective, the objective of integration is not to automate every coordination task. Some building problems are inherently interdisciplinary and require human judgment. Instead, coordination theory helps the analysis distinguish between necessary coordination and coordination created primarily by fragmented information. If a building operator spends time identifying where information is stored or which vendor is responsible, that coordination burden may be reducible. If a complex fault requires several specialists to interpret interacting systems, coordination may remain valuable even in an integrated architecture.

Vendor dependency can be interpreted within this framework as a particular form of interorganizational coordination. External specialists may hold knowledge that is critical to

diagnosis or intervention. The question is whether the building organization has sufficient internal knowledge and information to coordinate these specialists effectively. Ecosystem theory and infrastructure theory therefore meet at this point: the installed technical base and the ecosystem of actors jointly determine how coordination is performed (Hanseth & Lyytinen, 2010; Adner, 2017; Jacobides et al., 2018).

This produces an important analytical distinction between dependency and interdependence. Dependency implies that the ability of one actor to act is constrained by another actor's control over a necessary resource. Interdependence describes a broader relationship in which multiple actors rely on complementary contributions. The empirical analysis will therefore examine when vendor involvement represents productive interdependence and when it represents an avoidable concentration of knowledge, access, or control.

3.5 Governance capability

The concepts above are brought together through the central analytical construct of governance capability. The term is intentionally defined for this thesis rather than presented as an established universal measure. As introduced in Section 1.4, governance capability refers to the ability of relevant building actors to access, interpret, coordinate around, decide upon, and act on information necessary for managing the building.

This definition builds on the combined implications of infrastructure, platform governance, data governance, and organizational coordination. Infrastructure theory directs attention to the technical conditions of access and evolution (Star & Ruhleder, 1996; Hanseth & Lyytinen, 2010). Platform governance highlights interfaces, modularity and architectural control (Yoo et al., 2010; Tiwana, 2014). Data governance highlights authority, accountability, and decision rights around data (Khatri & Brown, 2010). Organizational studies highlight the reciprocal relationship between technologies and work practices (Orlikowski, 1992; Barley, 1986; Leonardi, 2011). Governance capability is the analytical construct that connects these elements to the building organization's practical ability to manage its environment.

The definition deliberately contains several stages. Access is necessary because actors cannot use information they cannot obtain. Interpretation is necessary because technically correct information may remain meaningless without context. Coordination is necessary because responsibility and expertise may be distributed. Decision-making is necessary because information becomes organizationally relevant when it affects a choice. Action is necessary

because governance is not only about knowing what is happening but also about being able to respond appropriately.

This also means that governance capability should not be equated with technological sophistication. A highly complex system may provide limited governance capability if users cannot interpret it or if responsibility for responding is unclear. Conversely, a relatively simple system can be highly effective if it reliably provides relevant information to the actor who can act. This follows the sociomaterial understanding that technological properties become meaningful through their interaction with organizational practices (Orlikowski, 1992; Leonardi, 2011).

Governance capability is therefore used in the empirical analysis as a way of asking whether the infrastructure helps or constrains organizational action. It is not measured numerically. Instead, indicators of capability will include the actors' descriptions of information access, coordination effort, dependence on specialists, ability to make decisions, ability to intervene, and the usability of integrated information.

3.6 Visibility, interpretability and actionability

The central conceptual distinction of the thesis is between visibility, interpretability, and actionability. None of these terms is new. Each has an established use in research on data quality, decision support, and knowledge use. What this thesis adds is to combine them into one sequence and to give the final stage an organizational meaning. The distinction is introduced to make the relationship between building data and governance capability analytically precise.

The term actionability has a long history in organizational and information research. Argyris (1996) uses the concept of actionable knowledge to describe knowledge that actors can use to produce intended consequences in practice. In this sense, knowledge is not valuable merely because it is valid; it must also be usable by someone acting in a real situation. In data mining, Cao (2010) makes a similar argument. He shows that patterns can be technically interesting without being actionable for the organization that is meant to use them, and he argues that actionability depends on domain context and on the decisions the knowledge is meant to support. Research on data quality points in the same direction. Wang and Strong (1996) show that data consumers judge quality not only by accuracy but also by accessibility, by representational qualities such as interpretability, and by contextual qualities such as relevancy and timeliness. Finally, Endsley (1995) describes situation awareness as a staged process of perception, comprehension, and projection that comes before decision and action. Taken

together, these contributions support the idea that information becomes useful through several distinct steps, and that failure is possible at each step.

This thesis builds on these contributions but differs from them in one important respect. Argyris (1996) and Endsley (1995) mainly concern the knowledge or awareness of an individual actor, and Cao (2010) and Wang and Strong (1996) mainly concern the properties of data and analytical results. In a multi-tenant building, by contrast, the steps from data to action are often distributed across systems, vendors, and organizational roles. The person who can see a signal may not be the person who can interpret it, and the person who can interpret it may not have the authority to act. Actionability is therefore defined here as an organizational rather than only a cognitive or technical property.

Visibility concerns whether relevant information can be observed, retrieved, or made available to an actor. It corresponds to the accessibility dimension of data quality (Wang & Strong, 1996) and to the perception stage of situation awareness (Endsley, 1995). In this study, visibility is closely related to digital infrastructure and data governance, because access depends on interfaces, protocols, architecture, and decision rights (Star & Ruhleder, 1996; Khatri & Brown, 2010). A technically inaccessible dataset cannot support a decision regardless of its potential informational value.

Interpretability concerns whether the actor can understand what the information means in context. Wang and Strong (1996) treat interpretability as a representational dimension of data quality, and Endsley (1995) describes comprehension as a separate stage from perception. The semantic interoperability literature illustrates why technical exchange is insufficient for this step. Qiang et al. (2023) show that building ontologies differ in their structures and that semantic differences can impede interoperability and reuse. Di Biccari et al. (2022) similarly demonstrate that information interoperability requires appropriate modeling and transfer strategies. These findings support the idea that a data point can be visible while still requiring substantial cognitive or organizational work before it becomes meaningful.

Actionability concerns whether information reaches an actor who has the responsibility, authority, and practical means to act on it, in time and in a form that is relevant to the decision at hand. It draws on the contextual qualities of timeliness and relevancy (Wang & Strong, 1996) and on the idea that knowledge is valuable when it can be used to produce intended consequences (Argyris, 1996; Cao, 2010). The organizational elements of responsibility and authority are added in this thesis. A building manager may understand that a temperature is

abnormal but still be unable to respond if the responsible supplier is unclear or if the system cannot be controlled. Similarly, an alarm that arrives after the consequence has occurred may have lower operational value than an earlier warning. Actionability thus connects information with organizational capability.

The distinction is represented in Figure 1, which also situates each transition within the theoretical perspective that motivates it.

Figure 1. Conceptual model linking infrastructure configuration to governance capability

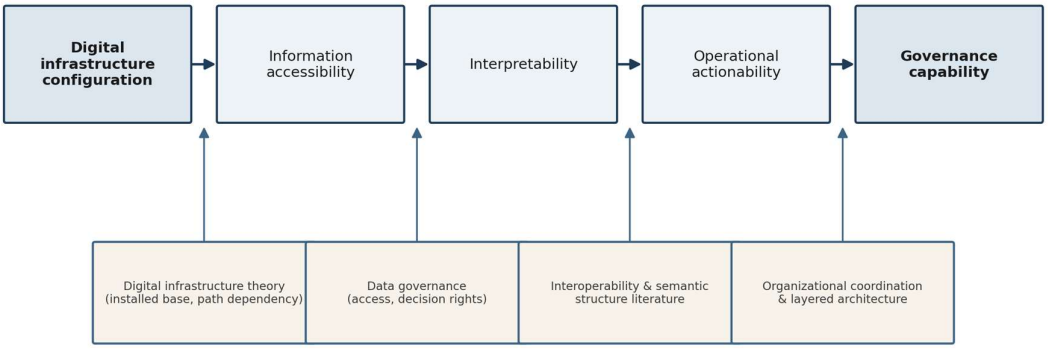


Figure 1. Conceptual model linking infrastructure configuration to governance capability, showing the theoretical perspective informing each transition.

Source: Author's own elaboration based on the literature reviewed in Chapters 2 and 3.

The sequence should not be understood as a deterministic pipeline. Failures can occur at any point. A system may be inaccessible. Data may be accessible but semantically unclear. Information may be interpretable but arrive too late. An actor may understand a problem but lack authority or technical ability to intervene. The value of integration therefore depends on maintaining the chain between information and action rather than maximizing any one technical attribute.

This distinction also helps explain why the thesis does not define successful integration as the creation of a single dashboard. A dashboard may increase visibility while leaving interpretation and action unchanged. Conversely, an integration layer may provide substantial value by exposing only a limited set of relevant signals if those signals are timely, understandable, and linked to action. The analytical focus is consequently on what the integrated infrastructure enables actors to do, not merely on what it allows them to see.

The propositions that follow from this framework are introduced in Section 8.1, where they are used together with a revised version of this conceptual model to structure the discussion of the empirical findings.

4. Methodology

This chapter describes how the study was designed and carried out. It explains the research design and its philosophical position, shows how the research questions, theoretical framework and methods are aligned, and describes case selection, data collection, the research process and the analysis. The chapter closes with a discussion of research quality, the strengths and limitations of the design, and ethical considerations.

4.1 Research design

The study adopts a qualitative single-case study design combining organizational inquiry, micro-ethnographic field observation, and design-oriented technical investigation. The choice of methodology follows directly from the research questions and the theoretical framework developed in the preceding chapters. The main research question asks how digital fragmentation of building infrastructure shapes governance capability in a multi-tenant commercial building. The first sub-question examines how fragmentation and infrastructural integration affect the distribution of information, knowledge, and coordination responsibilities, while the second examines the conditions under which integrated digital infrastructure can transform data visibility into operational actionability. These questions concern relationships, practices, perceptions, organizational responsibilities, and infrastructural conditions within a specific context. They therefore require a research design capable of examining the phenomenon in depth rather than isolating individual technical variables.

Bryman and Bell (2015) describe qualitative research as particularly concerned with understanding the meanings and perspectives through which social actors interpret their environments. This is relevant to the present study because governance capability cannot be established solely by measuring the technical availability of data. The study also needs to understand how different actors experience the infrastructure, which information they consider useful, how they coordinate with other actors, and where they experience dependencies or limitations. A qualitative approach therefore allows the research to investigate the organizational meaning of technical fragmentation rather than treating fragmentation solely as a measurable technical property.

The study uses a case-study strategy because the research phenomenon is closely connected to its context. Yin (2018) argues that case study research is appropriate when a contemporary phenomenon is investigated within its real-life context and when the boundary between the phenomenon and its context is difficult to separate. This describes the Majorstuhuset case

closely. Digital fragmentation cannot be isolated from the building's installed technical systems, organizational structure, vendor relationships, documentation, routines, and decision-making processes. The building is therefore treated as a socio-technical case in which infrastructure and organizational practice are mutually connected.

The design also contains a limited design-oriented technical component. Selected elements of an integration architecture were developed and tested in order to investigate how information could be accessed, processed, and presented without replacing the complete installed base. This component is not treated as a controlled experiment and does not attempt to establish a causal relationship between the prototype and improved building performance. Instead, technical investigation is used to expose practical infrastructural conditions. Some limitations are difficult to identify through interviews or documents alone and become visible only when an actor attempts to access, connect, configure, or reorganize an existing system.

The research design can therefore be characterized as a qualitative single-case study with a micro-ethnographic fieldwork component and a design-oriented technical investigation. The case study provides the overall research strategy, micro-ethnography provides situated empirical observation, interviews provide actor perspectives, and technical investigation provides direct knowledge of selected information pathways.

4.2 Philosophical positioning of the study

Following Bryman and Bell's (2015) framework for evaluating business research, the study's ontological and epistemological commitments are made explicit here because they shape what counts as acceptable evidence throughout the thesis.

Ontologically, the study adopts a constructionist rather than an objectivist position. Digital infrastructure is treated as a social product rather than a fixed technical fact: what counts as "the same" system, "fragmented" information, or a "governable" building depends on the practices and vocabularies of the actors who work with it (Star & Ruhleder, 1996; Orlikowski, 1992). This does not mean that technical properties are unimportant. Protocols, interfaces, and physical equipment have real effects that are not simply talked into being. The study therefore leans on a moderate, sociomaterial version of constructionism, consistent with Orlikowski and Scott's (2008) argument that technology, work, and organization are mutually implicated rather than either purely social or purely technical.

Epistemologically, the study is interpretivist. The research questions ask how actors experience, interpret, and act upon a fragmented digital environment, which requires access to meaning

rather than only to measurable technical states. Bryman and Bell (2015) contrast interpretivism with a positivist epistemology that treats the social world as amenable to the same causal methods used in the natural sciences; the present study instead seeks to understand governance capability from the perspective of the people who must exercise it. This is consistent with the choice of a qualitative single-case design, semi-structured interviewing, and reflexive thematic analysis described later in this chapter.

This positioning has three practical consequences for the research design. First, the researcher's own involvement in the technical investigation is treated as a source of situated knowledge to be made explicit and managed, rather than as contamination of an otherwise neutral measurement process (Section 4.5). Second, differences between participants' accounts are not treated as measurement error to be averaged away; they are treated as evidence that different actors occupy different positions within the same infrastructural arrangement (Section 4.7). Third, the analytical propositions developed in Section 8.1 are used as sensitizing devices rather than as hypotheses to be falsified, which is consistent with an interpretivist rather than a positivist logic of inquiry.

The study does not adopt a strict social-constructionist position in which technology has no independent effect, nor a strict realist position in which infrastructure determines organizational outcomes. Instead, it follows the sociomaterial middle ground already established in the theoretical framework (Orlikowski, 1992; Leonardi, 2011; Orlikowski & Scott, 2008): technical arrangements constrain and enable what actors can do, while actors' practices simultaneously give those arrangements their organizational meaning.

4.3 Consistency between research questions, theoretical framework, and methodology

A central methodological principle is consistency between research questions, theoretical framework, data collection, and analysis. The research questions investigate relationships between digital infrastructure and organizational governance. The theoretical framework combines digital infrastructure theory, layered architecture, ecosystem governance, data governance, and organizational coordination. Empirical methods were selected to provide evidence for these theoretical concepts. Table 1 summarizes this alignment.

Table 1. Alignment between research questions, theoretical perspectives, empirical sources and analytical purpose

Research focus	Theoretical perspective	Primary empirical source	Analytical purpose
Main RQ: fragmentation and governance	Infrastructure theory; governance capability	Four interviews, observation, system mapping	Examine how infrastructure influences access, interpretation, coordination, and action
RQ1: information and responsibilities	Infrastructure, ecosystem, platform governance	Role-specific interviews, observation	Examine how responsibilities and knowledge are distributed
RQ2: visibility and actionability	Data governance, layered architecture	Technical investigation, interviews	Identify conditions for data visibility to become useful action

Source: Author's own elaboration.

Each theoretical perspective directs attention toward specific analytical purposes. Digital infrastructure theory highlights the installed base, heterogeneity, and path dependency (Star & Ruhleder, 1996; Hanseth & Lyytinen, 2010). Layered architecture shows how capabilities can be added without replacing underlying components (Yoo et al., 2010). Ecosystem and platform perspectives highlight interdependence, interfaces, and modularity (Adner, 2017; Tiwana, 2014). Data governance directs attention to access rights and accountability (Khatri & Brown, 2010). Organizational theories of technology examine the interaction between technical systems and human practices (Barley, 1986; Leonardi, 2011).

This alignment ensures that research questions, theories, and methods form an analytical chain: the questions define what needs understanding; theory provides concepts for interpretation; methods provide access to relevant material; analysis connects observations back to the questions.

4.4 Case selection

Majorstuhuset was selected as a purposive case because it contains several characteristics directly relevant to the research problem. It is an existing multi-tenant commercial building

rather than a greenfield digital development. Its technical environment contains systems from different technological generations, several external actors, and multiple sources of operational information. Digitalization has also developed incrementally rather than according to one comprehensive architecture.

This makes the case particularly appropriate for examining the concept of the installed base introduced in Chapter 3. Existing technical systems cannot simply be assumed to disappear when new digital services are introduced. Instead, new technologies have to coexist with legacy equipment, established routines, vendor relationships, and existing information structures. Majorstuhuset therefore provides an empirical setting in which the tension between infrastructural continuity and digital reconfiguration can be observed.

The case also provides access to different actor positions. The facilitator is associated with strategic and governance concerns; the operations manager with day-to-day building management; the external integration vendor with technical interoperability and legacy systems; and the site engineer with technical development, documentation, and coordination. This diversity is important because the research questions explicitly concern the distribution of information, knowledge, and responsibilities. The same infrastructure can create different opportunities and constraints depending on the actor interacting with it.

The study does not claim that Majorstuhuset is statistically representative of commercial buildings more generally. The purpose of the case is analytical rather than statistical. Yin (2018) distinguishes case-study generalization from statistical generalization: the aim is to develop or examine theoretical explanations that may be relevant under comparable conditions. The transferability of the findings is therefore dependent on contextual similarities rather than numerical representativeness.

4.5 Researcher position and access

The researcher had direct practical access to Majorstuhuset and participated in the technical investigation of the building's infrastructure. This included examining the building network context, physical equipment, digital interfaces, available data sources, the ThinkEdge SE10, Node-RED, Airthings data access, and integration possibilities. Such access provided information that could not have been obtained entirely through interviews or documentary sources.

Researcher access is particularly important in the study of infrastructure because infrastructure is often embedded in routine activity and may become visible only when it fails or when

someone attempts to modify it. Star and Ruhleder (1996) emphasize the relational and situated character of infrastructure, while Orlikowski (1992) highlights the interaction between technological properties and human practices. Direct engagement with the building therefore helped reveal the difference between how infrastructure was formally described and how it could practically be accessed or used.

At the same time, the researcher's involvement creates a potential source of bias. The researcher was not a detached external observer and had a practical interest in investigating integration possibilities. This could create a tendency to interpret integration as desirable or to give greater attention to evidence supporting the project direction. The study addresses this risk through several measures.

First, integration is treated as a research question rather than an assumed solution. The analysis explicitly considers situations in which greater integration may create additional complexity, dependency, security concerns, or responsibility. Second, four actors with different positions were interviewed so that the researcher's technical perspective could be compared against strategic, operational, engineering, and vendor perspectives. Third, the analysis distinguishes between direct observation, participant accounts, documentary information, technical test results, and proposed architecture.

This distinction is essential for maintaining analytical discipline. A project document describing a planned system is not treated as evidence that the system was implemented. An interviewee's description of a legacy system is not presented as independently verified technical fact unless another source confirms it. A prototype demonstrating data access is not interpreted as evidence of improved organizational performance. These distinctions help reduce the risk that familiarity with the case becomes a substitute for empirical verification.

4.6 Data collection

The study combines four principal forms of empirical material: semi-structured interviews, micro-ethnographic field observation, system mapping and documentary material, and infrastructural prototyping. The purpose of combining these sources was not simply to increase the quantity of information collected. Each method provides a different perspective on the research problem.

4.6.1 Semi-structured interviews

Four semi-structured interviews were conducted with actors occupying complementary positions in the Majorstuhuset environment: a building facilitator or senior decision-maker, an operations manager, an external building-systems integration specialist, and a site engineer involved in development and digitalization. The participants were selected purposively because they possess different forms of knowledge, responsibility, and involvement with the infrastructure. This corresponds with Bryman and Bell's (2015) discussion of qualitative sampling, where participants may be selected because of their relevance to the research questions rather than because they statistically represent a wider population. The interview guide is presented in Appendix A and an overview of the interviews in Appendix B.

The interviews were structured around common themes while allowing the wording and emphasis to be adapted to the participant's role. Core themes included digital infrastructure, fragmentation, data accessibility, coordination, responsibility, vendor relationships, documentation, information interpretation, actionability, future digitalization, and the perceived value of integration.

The role-specific structure was deliberate. The facilitator was asked primarily about strategic information requirements, decision-making, ownership, investment, governance, and vendor relationships. The operations manager was asked about daily operation, failures, communication, alarms, historical data, and coordination with service providers. The integration vendor was asked about legacy systems, protocols, data access, interfaces, and practical integration work. The site engineer was asked about infrastructure development, documentation, semantic harmonization, contractor knowledge, and future digitalization.

The underlying themes therefore remained consistent, allowing comparison across interviews, while the questions were adapted so that participants could speak from their actual positions within the system. This is particularly relevant to RQ1, since the distribution of information and responsibility cannot be understood adequately from one organizational perspective.

4.6.2 Micro-ethnographic field observation

A micro-ethnographic fieldwork component was used to supplement the interviews by bringing attention to situated activities, material arrangements, interactions, and practices within the building. Alvehus and Crevani (2022) present micro-ethnography as an approach for studying organizational activity closely and in situ, with particular attention to how practices emerge through interaction and multimodal forms of activity.

The term is used in this thesis in a focused sense. The study does not claim to constitute a long-term ethnographic investigation of Majorstuhuset's organizational culture. Instead, micro-ethnography refers to the close observation of particular activities and settings relevant to the research problem: the building's technical environment, systems, interfaces, equipment, information pathways, and the practical work of investigating and navigating them.

During site visits and technical activities, attention was given to how systems were physically arranged, where information was available, how information had to be located or traced, how building personnel encountered different systems, which systems required specialist knowledge, and how technical boundaries became visible during practical work.

This method is particularly useful because some infrastructural characteristics are routine and therefore may not be articulated by participants. An actor may describe information as "available" because they already know the procedure for accessing it. Direct observation can reveal that accessing the information requires several steps, multiple interfaces, or tacit knowledge. Micro-ethnographic observation therefore contributes to the distinction between formal availability and practical accessibility, which is central to the governance-capability concept.

Field observations were recorded through notes and used as empirical material alongside interviews, system mapping, documents, and technical investigation. They were treated as situated observations rather than neutral measurements. Their credibility comes from proximity to the setting and from comparison with interviews, documents, and technical tests rather than from an assumption that the researcher has an objective view of the organization.

4.6.3 System mapping and documentary material

System mapping was used to establish the structural relationships among relevant systems, information sources, interfaces, and actors. The mapping focused on four questions: where information originates, how it moves, where it is processed, and who or what mediates access to it.

This mapping was especially important for connecting the empirical case to the theoretical concepts of digital infrastructure and layered architecture. It allowed the research to identify not only individual systems but also the boundaries between them.

Documentary material included available technical descriptions, system inventories, project documentation, configuration information, and architectural proposals. Documents were used

to establish the documented or intended state of the infrastructure. However, they were not automatically treated as evidence of current operational capability. Existing infrastructure can change over time, and project documentation can describe planned states that were never fully implemented.

The distinction between documented architecture and observed architecture is therefore maintained throughout the thesis.

4.6.4 Infrastructural prototyping

The technical investigation included limited infrastructural prototyping designed to explore selected information pathways. A Lenovo ThinkEdge SE10 was used as a building-side computing point, and Node-RED was used for data processing and dashboard development. An Airthings source was accessed through its cloud API, providing a concrete example of how information from an existing service could be brought into a locally organized processing environment.

The confirmed Airthings pathway ran from the Airthings cloud through its REST API to the ThinkEdge device, into Node-RED for flow processing, and finally to the Node-RED dashboard. This demonstrated that information from an existing digital service could be retrieved and reorganized within a building-side processing environment without modifying the physical sensor system.

The prototype is therefore treated as evidence of technical accessibility and infrastructural reconfiguration, not as evidence that the building achieved measurable improvements in energy performance, maintenance, or governance.

Other elements of the broader architectural concept, including MQTT, ESP32 nodes, additional protocol adapters, and wider automation functions, were explored as part of the project direction but were not all implemented as one completed building-wide system. The external vendor's BACnet/Modbus integration activity was treated as empirical evidence of another actor's technical practice and not represented as the researcher's own implementation. Appendix E summarizes the implementation status of each component.

4.7 Integration of methods and methodological triangulation

The methods were deliberately combined because the research problem exists simultaneously at technical, organizational, and situated levels. Technical system mapping can show that two systems are separate, but it cannot fully explain the organizational consequences of that

separation. Interviews can reveal that an actor experiences information as difficult to use, but participants may not describe routine material details that are obvious to them. Observation can reveal these situated practices, while technical investigation can expose constraints that only become visible when someone attempts to cross a system boundary.

The study therefore uses methodological triangulation. Baxter and Jack (2008) emphasize the value of multiple sources of evidence in qualitative case study research, while Bryman and Bell (2015) discuss the importance of using suitable methods to address the particular research question rather than selecting methods independently of the research problem.

Triangulation is not interpreted as requiring all sources to agree. Differences between sources can themselves be analytically valuable. For example, an integration vendor may describe BACnet communication as technically straightforward, while an operations manager may still experience the overall process of obtaining useful information as cumbersome. These statements are not necessarily contradictory. They may reveal different levels of the same infrastructure: technical accessibility at one level and operational actionability at another.

The methods therefore complement rather than simply duplicate one another:
• Interviews provide actor perspectives, experiences, responsibilities, and interpretations.
• Micro-ethnographic observation provides situated evidence of practices and material arrangements.
• System mapping and documents provide structural and formal representations of the infrastructure.
• Technical investigation provides direct evidence of selected information pathways and integration possibilities.

This combination also supports the conservative causal position of the thesis. The study does not infer that integration improves governance simply because technical integration is possible. Instead, the different methods establish different parts of the argument, which are brought together during analysis.

4.8 Research process

The research process was iterative rather than strictly linear. The study began with an initial exploration of the building and its digital environment. System mapping and field observation provided an initial understanding of the installed infrastructure and highlighted areas where information and responsibility appeared fragmented.

This initial understanding informed the development of the interview themes. The interviews then provided additional perspectives on how the building was actually operated, which information actors considered important, how problems were coordinated, and where vendor dependency appeared.

Technical investigation was conducted alongside this organizational inquiry. Attempts to access, process, and display selected information sources revealed practical constraints that could not have been identified solely through conceptual system mapping. These experiences also informed the refinement of the conceptual distinction between information visibility, interpretability, and actionability.

The research process can therefore be summarized as the eight stages shown in Table 2.

Table 2. Stages of the research process

Stage	Activity
1	Initial system mapping
2	Field observation
3	Stakeholder inquiry
4	Identification of infrastructural and governance issues
5	Targeted technical investigation
6	Cross-source comparison
7	Thematic analysis
8	Theoretical interpretation

Source: Author's own elaboration.

This iterative process is consistent with the character of qualitative research described by Bryman and Bell (2015), where the relationship between data collection, theory, and analysis may develop during the research process rather than following a completely predetermined sequence.

4.9 Data analysis

The empirical material was analyzed using thematic analysis. Braun and Clarke (2006) describe thematic analysis as a flexible approach to identifying and interpreting patterns across qualitative datasets. Their later work emphasizes that reflexive thematic analysis involves active researcher interpretation and should not be presented as a mechanical process through which objectively existing themes are simply "discovered" (Braun & Clarke, 2021). This study follows that interpretive position and treats coding as a structured but reflexive analytical process.

The analysis therefore followed a structured but reflexive procedure. The researcher first became familiar with the interview transcripts, field notes, system maps, documents, and technical material. Initial codes were then developed around recurring topics and concepts relevant to the research questions.

The initial coding categories included:

- infrastructure;
- fragmentation;
- information access;
- information quality;
- documentation;
- interpretation;
- coordination;
- responsibility;
- vendor dependency;
- alarms;
- timeliness;
- semantic harmonization;
- integration;
- control;
- security;
- decision-making; and
- actionability.

Related codes were grouped into broader themes and subsequently compared across actor positions. The purpose of cross-actor comparison was to identify both common patterns and meaningful differences. The grouping of initial codes into themes is shown in Appendix C.

The analysis was theory-informed but not theory-determined. The theoretical framework sensitized the researcher to concepts such as installed bases, modularity, interdependence, governance, dependency, and decision rights. At the same time, empirical observations were allowed to modify and refine the analytical categories. The distinction between data visibility and actionability, for example, developed through the comparison of actors' different accounts of what constitutes useful information.

The analysis also deliberately separates description, finding, and interpretation. Chapter 5 presents the empirical context descriptively, with the detailed description of the infrastructure given in Appendix D. Chapter 6 presents patterns identified in the empirical material. Chapter 7 interprets those patterns through the theoretical framework. This distinction reduces the risk of presenting a theoretical interpretation as if it were directly stated by a participant.

4.10 Research quality: validity, reliability, and generalizability

Bryman and Bell (2015) devote substantial attention to the criteria used to evaluate business research and discuss the longstanding debate concerning how conventional concepts such as reliability and validity should be applied to qualitative research. They note that qualitative researchers have proposed alternative criteria rather than simply transferring quantitative standards unchanged. A key alternative is Lincoln and Guba's (1985) concept of trustworthiness, which includes credibility, transferability, dependability, and confirmability. In this study, these qualitative criteria are used alongside the teacher's requested concepts of validity, reliability, and generalizability.

4.10.1 Validity and credibility

Validity concerns whether the study provides a credible account of the phenomenon it claims to investigate. For this study, validity begins with conceptual consistency between the research questions, theoretical framework, empirical material, and analysis.

Construct validity is strengthened by linking each central construct to identifiable empirical evidence. Digital fragmentation is examined through system mapping, technical observation, documentary material, and actor accounts. Governance capability is examined through information access, interpretation, coordination, decision-making, and ability to act. Operational actionability is examined through participants' accounts of information relevance, timeliness, alarms, interpretation, responsibility, and intervention.

The conceptual model developed in Chapter 3 (Figure 1) also functions as a consistency check, linking infrastructure configuration to information accessibility, interpretability, operational actionability, and ultimately governance capability.

The model is not treated as an empirically proven causal chain. Instead, it provides an analytical structure through which the case evidence can be examined. The purpose is to determine where the empirical material supports, complicates, or challenges these relationships.

Credibility is further strengthened through triangulation. The study does not depend on a single interview or one technical observation. The four participants occupy different positions, and their accounts are compared with field observation, documents, system mapping, and technical investigation. Where different sources provide different accounts, the difference is not automatically treated as an error. It is considered as a possible indication that different actors occupy different positions within the same infrastructural arrangement.

The study also applies a conservative standard when making empirical claims. Participant statements are treated as participant perspectives unless independently corroborated. Technical tests are reported only for systems that were actually accessed or tested. Proposed architecture is clearly distinguished from implemented architecture. The prototype is not used to claim measured improvements in governance or energy performance.

4.10.2 Reliability and dependability

Reliability is more difficult to interpret in qualitative research because qualitative analysis is not intended to produce identical measurements each time a researcher encounters a social phenomenon. Bryman and Bell (2015) discuss this limitation and note that qualitative researchers have proposed alternative ways of considering reliability, particularly through transparency and dependability.

For the present study, reliability is therefore approached primarily as dependability and traceability of the research process. The research maintains recorded interview material and transcripts, a documented interview structure, a coding framework, field notes, system maps, and technical records. The process through which evidence was transformed into findings is explicitly described.

The study also distinguishes between procedural consistency and interpretive identity. Another researcher may not necessarily produce identical interpretations of the Majorstuhuset case, particularly given the interpretive nature of the study. However, another researcher should be

able to understand how the present interpretations were developed and which empirical material supports them. This is the form of reliability most appropriate to the present research design.

Reflexivity is part of this process. Braun and Clarke (2021) argue that reflexive thematic analysis involves researcher interpretation and that methodological quality should not be established by pretending that the researcher is absent. Instead, rigour is associated with transparency, coherence, and sustained engagement with the data. The researcher's practical involvement is therefore treated as a methodological condition that needs to be acknowledged and managed rather than hidden.

4.10.3 Generalizability and transferability

The study does not support statistical generalization from Majorstuhuset to all multi-tenant commercial buildings. The case was purposively selected and the interview sample was not designed to statistically represent a population. Applying numerical generalization would therefore be inappropriate.

Bryman and Bell (2015) discuss the problem of generalization in qualitative research and distinguish the logic of qualitative transferability from statistical generalization. Lincoln and Guba (1985) similarly use transferability to describe the extent to which insights may be applicable in other contexts when sufficient contextual information is provided.

The contribution of this thesis is therefore analytical. It identifies mechanisms and conditions through which infrastructural fragmentation may affect information, knowledge, coordination, and action. These mechanisms may be relevant to other existing buildings that share characteristics such as heterogeneous installed systems, multiple vendors, distributed knowledge, and incremental digitalization.

Transferability is supported by providing detailed contextual information about Majorstuhuset, its actors, infrastructure, methods, and limitations. The reader is thereby given sufficient information to assess whether the findings are relevant to another case. The thesis does not claim that every commercial building will experience fragmentation in exactly the same way.

4.10.4 Confirmability and researcher reflexivity

Confirmability concerns the extent to which findings are grounded in the empirical material rather than being primarily products of the researcher's preferences or assumptions. This is particularly relevant because the researcher participated in the technical investigation and had an interest in exploring infrastructural integration.

The study addresses this by maintaining a distinction between what was observed, what participants reported, what documents stated, and what the researcher inferred. Where the evidence is incomplete, the analysis acknowledges the uncertainty rather than filling the gap with assumption.

The study also deliberately includes evidence that complicates the integration proposition. Integration is not treated as universally beneficial. The analysis considers potential increases in complexity, security requirements, new dependencies, specialist knowledge requirements, and organizational responsibilities. This provides a counterweight to the researcher's practical involvement in developing an integration architecture.

The aim is therefore not to claim complete objectivity. Instead, confirmability is supported by transparent evidence classification, traceable analytical decisions, triangulation, and explicit reflexivity.

4.11 Strengths of the research design

The first strength of the design is its multi-perspective character. Four interviews provide different views of the same infrastructural environment. This is particularly important because the research questions concern distribution and interdependence. A facilitator, operations manager, vendor, and site engineer encounter different parts of the same system.

The second strength is the combination of organizational and technical evidence. Interview data provides information about actor experiences and responsibilities, while technical investigation provides evidence about actual information pathways. This helps prevent the analysis from becoming either purely technical or purely perceptual.

The third strength is the use of micro-ethnographic field observation. Observation provides access to situated practices and material arrangements that may not be articulated in interviews. This is particularly useful for an infrastructure study because many aspects of an installed system become visible through interaction with it rather than through formal descriptions.

The fourth strength is the iterative relationship between theory and empirical investigation. Rather than treating the technical work and interviews as separate activities, the research allowed observations from one stage to inform subsequent questions and investigation. This supports the development of a more empirically grounded conceptual model.

The fifth strength is the clear distinction between implemented, observed, and proposed architecture. This strengthens the credibility of the technical findings because the study does not claim outcomes that were not actually demonstrated.

4.12 Limitations of the research design

The principal limitation is the use of a single case. The design provides depth but limits the breadth of comparison. Majorstuhuset offers a particularly useful environment for studying fragmentation, but other buildings may have different ownership structures, contractual relationships, technical architectures, or organizational capabilities.

The second limitation is the limited number of interviews. The four participants represent strategically selected positions, but they cannot represent every stakeholder in the building. In particular, the study does not claim to represent all tenant experiences. The selection is instead justified by the need to examine the actors most directly involved in building governance, operations, integration, and engineering.

The third limitation is the incomplete state of the broader integration architecture. Selected information pathways were implemented or tested, but the entire proposed building-wide architecture was not deployed as a production system. This means that the thesis can examine technical feasibility and organizational conditions but cannot demonstrate long-term performance improvements.

The fourth limitation concerns access to proprietary and legacy systems. Some technical information could only be accessed through existing interfaces, external specialists, documentation, or participant descriptions. Where direct verification was impossible, the thesis treats the information as bounded evidence rather than as definitive technical fact.

A fifth limitation concerns researcher involvement. Practical participation provided valuable access but also increased the possibility of confirmation bias. The study therefore relies on triangulation, reflexivity, and clear evidence classification to reduce this risk.

Finally, the study focuses on the relationship between infrastructure and governance and does not provide a full quantitative evaluation of energy performance, cost savings, cybersecurity, or organizational productivity. These dimensions are relevant to future research but lie outside the empirical scope of this study.

4.13 Ethical considerations

The interviews were conducted for academic research purposes, and participants were informed about the purpose of the study and their participation. The analysis focuses primarily on organizational roles, infrastructural arrangements, practices, and system relationships rather than unnecessary personal attribution.

Because the case involves relationships between the building organization and external suppliers, particular care is required when discussing weaknesses, dependencies, or service issues. The purpose of the analysis is not to assign blame to individual actors or companies, but to understand how infrastructural and organizational arrangements affect coordination and governance.

Technical information was handled with particular attention to security and confidentiality. Credentials, access tokens, private network information, passwords, and other sensitive technical details are excluded from the thesis. The architecture is described at a level sufficient for academic analysis without exposing information that could facilitate unauthorized access.

The ethical approach therefore reflects the same principle that guides the broader research: information should be made sufficiently available for legitimate understanding and analysis without assuming that unrestricted access is appropriate.

5. Majorstuhuset: Case and Empirical Context

This chapter introduces the case. It describes the building and its operational context and the actors involved in its operation and development, and summarizes the main characteristics of the building's digital environment that are relevant to the findings.

5.1 Building and operational context

Majorstuhuset is an established multi-tenant commercial building in Oslo. Unlike a greenfield smart-building development in which the technical architecture can be defined before installation, the digital environment examined in this study has developed incrementally. Its present condition is the result of earlier installations, replacements, operational requirements, supplier relationships, and subsequent attempts to introduce new digital capabilities. The building is therefore best understood as an existing installed base in which systems from different periods have to coexist.

This characteristic is important for the empirical case because the practical challenge is not simply to design an ideal digital architecture from the beginning. The challenge is to improve information coherence while retaining existing assets, minimizing disruption, and working with systems that differ in technical age, accessibility, and architectural openness. The initial problem formulation of the study described the building environment as fragmented, with heterogeneous systems and multiple actors.

The building's operational setting is also multi-tenant. This means that the digital infrastructure of interest in this study is not intended to replace or absorb the private networks of individual tenants. The focus is instead on the building-side digital infrastructure used to support the management and coordination of building systems and information. This distinction matters because the case is not about creating one common network for every organization operating in the building. It is about creating a coherent layer for building-owned information and services while preserving appropriate separation between organizational domains.

The empirical material consequently places the building between two conditions. On one side, the building already contains substantial digital capability: sensors, building-management functions, energy-related information, cloud-connected services, controllers, and digital interfaces. On the other side, these capabilities are distributed across different technical and organizational arrangements. The case therefore offers an opportunity to examine a common digitalization paradox: an environment can be information-rich while remaining difficult to govern as a whole.

The practical significance of this setting is also reflected in the project's evolving analytical direction. Earlier project assumptions placed stronger emphasis on technical integration and interoperability. As the research developed, the focus shifted toward the organizational consequences of fragmented infrastructure and the conditions under which digital visibility can become useful for governance. The study therefore moved from a relatively technical investigation toward a broader Industrial Management and Innovation discussion concerning digital infrastructures, governance capability, and platform-oriented ecosystems.

5.2 Actor landscape

The case involves several actor positions that contribute to the operation, interpretation, and development of the building's digital environment. At a strategic level, building management and facilitation are responsible for decisions concerning the direction of the property, investment, priorities, and relationships with external service providers. At an operational level, personnel are responsible for responding to problems, interacting with tenants, monitoring conditions, and coordinating technical interventions. External vendors possess specialist knowledge about individual building systems. Engineers and integration specialists work across technical boundaries and contribute to development and reconfiguration. Tenants experience the consequences of technical performance, although they do not necessarily control the building-owned infrastructure.

These actors do not possess the same knowledge or require the same information. The facilitator's perspective is primarily concerned with what information is necessary for strategic decisions and whether a digital investment creates practical value. The operations manager works closer to the physical building and therefore needs information about equipment states, environmental conditions, alarms, historical behavior, and problems requiring intervention. The external integration vendor approaches the building through protocols, controllers, interfaces, and system-specific technical knowledge. The site engineer works across these boundaries and is concerned with system development, documentation, semantic consistency, and coordination.

The result is a distributed knowledge environment. Knowledge about the building is not located in a single repository or held by a single actor. Some knowledge is embedded in the building organization's operational experience; some is associated with specific suppliers; some exists in technical documentation; and some must be reconstructed by physically tracing equipment or interfaces. The actor landscape is therefore relevant to the research questions because the

distribution of actors corresponds with the distribution of information, expertise, responsibility, and decision-making capacity.

The empirical work also indicates that these actors are interdependent without being organized under a single hierarchy. The building organization may require external specialist expertise, while suppliers require access to the building context and information to perform their work. The integration specialist may be able to connect systems technically but does not control strategic decisions about the building. The same infrastructure therefore links actors who have different forms of authority and different purposes. The building's digital environment can therefore be described as an infrastructural ecosystem characterized by layers of technologies and distributed governance rather than as a single automation system.

5.3 Empirical context summary

A detailed description of the building's existing digital infrastructure, its information flows, the ThinkEdge intervention, the Airthings integration and the boundaries of legacy-system integration is provided in Appendix D. The main characteristics of the case are summarized below.

Majorstuhuset can therefore be characterized as an existing, layered, multi-actor digital environment rather than as a single smart-building platform. The building contains significant digital capability, but that capability is distributed across different systems, vendors, interfaces, and forms of expertise. The resulting fragmentation is partly technical, partly informational, and partly organizational.

The empirical context also reveals that integration is already occurring, but not as a complete transformation. Different approaches coexist: cloud/API-based access through Airthings, existing building-management platforms such as Energy Control, legacy-system integration through protocols such as BACnet and Modbus, and the exploration of a building-owned edge layer based on the ThinkEdge and Node-RED.

This makes Majorstuhuset a useful case for the research questions because it contains both sides of the phenomenon under investigation. Fragmentation creates distributed information, knowledge, and coordination requirements. At the same time, integration activities demonstrate possible mechanisms for reducing some of these burdens. The central empirical question is therefore not whether integration is technically possible. It is what happens to information, responsibility, and action when integration is introduced into an already fragmented environment.

The next chapter presents the findings from the four interviews and fieldwork in thematic form. The analysis then uses these findings to examine how the infrastructural conditions described here influence governance capability.

6. Findings

This chapter presents the empirical findings from the four interviews, micro-ethnographic field observations, documentary material, system mapping, and technical investigation described in Chapter 4. The chapter is intentionally descriptive and thematic. It does not introduce new theoretical concepts or provide the main theoretical interpretation; those tasks are reserved for Chapter 7. The purpose here is to show the empirical patterns that emerged from the material and to establish the evidence base from which the later analysis develops.

The findings are organized around six recurring themes that emerged across the empirical material: distributed information and knowledge, distributed coordination responsibility, differentiated information requirements, the continuing role of vendor expertise, the distinction between data visibility and actionability, and the importance of semantic harmonization during integration. The themes are not treated as separate problems. They are presented as connected aspects of the same empirical environment.

6.1 Fragmented information and distributed knowledge

Across the interviews, fragmentation appeared less as a complete absence of information than as a situation in which information and the knowledge required to use it were distributed across several systems and actors. Different participants encountered different parts of the building's information environment according to their roles. The facilitator was primarily concerned with the information needed for strategic decisions and oversight. The operations manager worked closer to day-to-day building conditions and recurring technical problems. The external integration vendor approached the building through controllers, communication protocols, tags, and system interfaces. The site engineer considered the relationships between systems, documentation, and the structure of information across technical domains.

The interviews therefore did not describe one common information space in which all actors had the same view of the building. Instead, the information environment was differentiated by system boundaries and by organizational roles. A technical condition could be represented in one platform while the practical knowledge needed to interpret that condition remained with an experienced operator or an external specialist. The existence of information and the existence of knowledge about that information were therefore not the same thing.

The operations perspective illustrates this particularly clearly. Knowledge about building operation is partly accumulated through recurring incidents and previous interactions with service companies. Knowing what happened before and which company normally deals with a

particular problem provides a form of operational knowledge that is not necessarily contained in the technical systems themselves. Similarly, the external integration vendor's work depended not simply on discovering that a controller communicated through a particular protocol, but on understanding the controller environment, available tags, system configuration, and the physical context in which the information was produced.

The site engineer's emphasis on documentation adds another dimension. Where system information is poorly documented or remains concentrated with contractors, the building organization may have access to parts of its infrastructure without possessing a complete understanding of it. This was described as an issue because building systems are long-lived and can change personnel and suppliers over time. In such circumstances, organizational knowledge can become fragmented even when individual systems continue to operate normally.

The facilitator described this distribution directly when reflecting on how many parties are currently involved in the building's climate system:

> "It is quite a big team now. There are many people involved. I think we counted around 12 different vendors who are, in some way, involved in the climate system. […] Everybody knows a little about the overall system, or they know a lot about their specific area, but they don't really know much about the other areas. So we need a guide." (Facilitator interview, personal communication, 2026)

This observation captures the empirical core of the theme: the fragmentation experienced at Majorstuhuset is not primarily a shortage of technical systems, but a situation in which no single actor holds a complete picture, and each vendor's knowledge stops at the boundary of its own subsystem. The site engineer described the practical consequence of this boundary when trying to reconstruct the building's history for the digital-twin project: "we had to spend enormous amounts of time around that documentation because we didn't know exactly how to name those walls… getting to that information took like two or three days for me" (Site Engineer interview, personal communication, 2026).

Taken together, the evidence suggests that fragmentation in Majorstuhuset has an informational and a knowledge dimension. Information can be distributed across platforms, while knowledge about how to interpret, locate, and use that information can be distributed among people. The two forms of fragmentation reinforce one another: the more difficult it is to establish a common information environment, the more important individual experience and specialist knowledge become in connecting the pieces.

6.2 Fragmentation distributes coordination responsibility

The second recurring finding concerns the amount of coordination required to move from a technical problem to an operational response. The operations manager described a process in which issues may first be discovered through observation or through a problem becoming noticeable in the building. Information is then communicated, often through telephone or SMS, and the likely responsible service provider is identified. When the situation is straightforward, this can be efficient because the relevant contact is already known. When responsibility crosses system or vendor boundaries, however, additional coordination becomes necessary.

The empirical process follows the same six steps shown in Table 3: problem detection, information communication, identification of the responsible actor, vendor contact, diagnosis, and coordinated intervention.

Table 3. Problem-to-intervention process at Majorstuhuset

Step	Activity
1	Problem detected
2	Information communicated
3	Responsible actor identified
4	Vendor contacted
5	Diagnosis
6	Intervention coordinated

Source: Author's own elaboration based on interview data.

Each stage contains an organizational task. Someone has to recognize that a condition requires attention. Someone must describe the problem sufficiently for another actor to understand it. Someone has to identify who is responsible for the relevant technical system. The specialist then has to diagnose the problem, after which the building organization coordinates the intervention. None of these activities necessarily appears in the technical architecture itself, yet they are part of the practical process through which the building is operated.

The interviews indicate that experience plays an important role in making this process manageable. Personnel who know the building and its suppliers can often shorten the route from detection to action because they already know where to direct the issue. This knowledge acts as

an informal coordination mechanism. However, the same dependence can become a weakness when information is not documented or when the person holding the relevant knowledge is unavailable.

The operations manager described a concrete case in which responsibility for a fault could not be settled without cross-vendor coordination:

> "There have been cases where we have contacted the ventilation company, but then they have had to contact those who have the heating system or the cooling system. There are different vendors, and in some cases the responsibility actually falls between the vendors." (Operations Manager interview, personal communication, September 7, 2026)

The facilitator described the same problem from a strategic vantage point, framing the building's need not as more data but as a coordinating role capable of connecting the specialist knowledge already present among the vendors: "We need a guide. We need someone to lead us through the different systems, and we need more knowledge ourselves to be able to judge what is good and what is bad" (Facilitator interview, personal communication, 2026).

The findings also show that manual coordination should not automatically be treated as a failure. Participants did not suggest that every interaction with a supplier should disappear. Some technical problems require specialist diagnosis, and some situations are too context-dependent for a fully automated response. The empirical issue is instead the amount of coordination that is generated by fragmented information. Where several actors have to be contacted simply to establish where information resides or who is responsible, coordination is experienced as an additional burden.

The distribution of coordination responsibility is therefore closely connected to the distribution of information and knowledge. When systems do not provide a shared operational picture, people become the connection points between systems. They route information, establish responsibility, and bring different forms of expertise together.

6.3 Different actors need different information

A third finding concerns the different information requirements of the actors involved in building operation. The interviews did not support the idea that all users should receive the same amount or type of information. Instead, information requirements differed according to responsibility, technical competence, and the decisions each actor was expected to make.

The facilitator's perspective emphasized relevance and the principle that information should be available when needed rather than requiring continuous inspection of detailed technical systems, a position summarized directly: "I don't need data that I don't need. I just need to know what I need to know, on a need-to-know basis" (Facilitator interview, personal communication, 2026). From this perspective, the usefulness of a digital interface depends partly on its ability to distinguish important conditions from the large amount of information that modern building systems can generate.

The operations manager expressed a different requirement. Historical information and current values can be useful for understanding recurring problems, following temperature and ventilation conditions, and identifying changes in building performance. The operations perspective therefore requires more detail than the strategic perspective, but that detail is still connected to specific operational purposes.

The external integration vendor operates at a different level again. Effective diagnosis can require controller-level information, communication details, tags, and knowledge about individual systems. Information that appears excessive from a strategic perspective may be necessary for technical intervention. The site engineer adds yet another requirement, arguing that the same underlying value needs to be presented differently depending on who is reading it:

> "Not only delta T is this much, but delta T is this much and this means that it affects the system in this way. […] If you can cognitively create the understanding with the facilitator who is not technical, then I think it's a big opportunity of just every day improving a building piece by piece." (Site Engineer interview, personal communication, 2026)

Information must therefore be organized sufficiently well that relationships between systems and comparable values can be understood across the building, and translated into a register appropriate to each actor's technical background.

These differences were important in shaping the empirical understanding of integration. A common information infrastructure does not necessarily mean a common user interface. The underlying data can be organized coherently while being presented differently to different actors. The empirical material therefore supports the idea of differentiated access and representation rather than one universal dashboard designed to satisfy every user simultaneously.

The distinction also clarifies why the amount of data displayed is a poor measure of digitalization success. More information can become less useful when it increases the effort required to find what matters. Participants consistently described information in terms of what it enables them to do, not in terms of how much information they could technically obtain.

6.4 Vendor expertise remains necessary

The fourth finding concerns the role of external vendors. The empirical material does not support a simple interpretation in which vendor involvement is itself the main problem. On the contrary, participants described specialist vendors as necessary and often valuable because they possess technical knowledge that would be difficult or inefficient for the building organization to maintain internally.

The operations manager described relying on vendors when a problem requires diagnosis beyond what can be established internally. The reason is practical: a vendor familiar with a particular controller, ventilation system, or other subsystem can often identify the issue faster and provide technical advice based on previous experience. The site engineer similarly emphasized the importance of contractor competence and the value that specialist knowledge can bring to building development and maintenance.

The external integration vendor provided a different example of the same phenomenon. Legacy-system integration depends on knowledge of communication protocols, controller configuration, available data points, and relationships between digital values and physical equipment. Even when a system uses BACnet or Modbus, successful integration can therefore depend on someone who understands the specific implementation rather than only the standard itself.

At the same time, the interviews identified a risk associated with concentrated external knowledge. When documentation is incomplete or technical knowledge remains mainly with a contractor, the building organization may struggle to evaluate information independently or determine how a system should be changed. The facilitator articulated this risk directly when describing how investment decisions are currently made:

> "We trusted a third party with it. You don't necessarily know the system yourself, but we thought it looked fancy and nice. […] I guess many people in my position trust their technical officer. If it doesn't work, the technical officer may be replaced. So it is probably more beneficial to have enough knowledge yourself to be able to judge whether something is good or bad." (Facilitator interview, personal communication, 2026)

This does not mean that all knowledge must be internalized. Instead, the concern is whether the organization retains enough understanding to coordinate specialist expertise and make informed decisions about its infrastructure.

This distinction between necessary expertise and problematic dependency is important. The empirical material points toward interdependence rather than a simple choice between internal control and external outsourcing. Building operation requires specialist competence, but the organization also needs sufficient architectural knowledge, documentation, and access to avoid making one supplier the unavoidable source of understanding for critical building functions.

6.5 Data visibility does not automatically create actionability

The fifth finding emerged particularly strongly from the operational perspective. Participants recognized that building systems already provide substantial amounts of information, but they did not regard visibility as sufficient in itself. Information was valued when it could support timely intervention, prevent a problem from becoming more serious, or make a decision easier to take.

The operations manager gave examples related to temperature, ventilation, and potential water-leakage situations, framing the value of information explicitly in terms of timing and consequence:

> "If I have the information soon enough, then we can solve things before they cause damage. For instance, water leakage is a very big threat to the building. If you have a leakage, it is crucial that we have an early warning, preferably automatically, and then we have an alarm so we can act on it." (Operations Manager interview, personal communication, September 7, 2026)

A signal received early enough can allow personnel to investigate a problem before tenant comfort is affected or physical damage occurs. A signal received only after the consequence is visible has a different operational value.

This creates a distinction between a dashboard that contains information and a system that supports response. A historical temperature graph may be useful for investigating a previous incident. An alarm may be more useful for preventing a future incident. Both use the same underlying category of information but support different operational purposes.

The interviews also indicate that an alarm by itself is not automatically actionable. A useful alarm needs an understandable condition, an appropriate threshold or trigger, a relevant recipient, and some means of responding. If an alarm is sent to someone who does not know

what it means or does not have responsibility or authority to act, the existence of the alarm does not necessarily increase governance capability.

The facilitator's perspective reinforces this finding from a different direction, warning explicitly against the opposite failure mode: "it is very important when you build an architecture that you are conscious about what you do not need to know, because you really want to avoid information overload, which is very easy to create" (Facilitator interview, personal communication, 2026). A large quantity of information can create an interpretation problem if users are expected to determine for themselves which values matter. The desired outcome is therefore not maximum visibility but relevant visibility. Information has to be translated into a representation that allows the actor to understand its significance without losing access to deeper information when required.

The empirical material consequently supports a distinction between three conditions. First, information must be available. Second, it must be understandable in context. Third, it must be sufficiently connected to an action or decision. Participants described these conditions in practical rather than theoretical language: knowing when something is wrong, knowing what it means, and knowing what needs to happen next.

Actionability was therefore found to be role-dependent. The same temperature value may be useful to a technical specialist investigating a ventilation issue, less useful to a senior decision-maker without context, and highly useful to an operations manager when it is combined with a threshold and an appropriate response process. The operational usefulness of information thus depends on its relationship with the actor and task.

6.6 Integration and semantic harmonization

The sixth finding concerns what happens after information has been made technically accessible. The external vendor described the practical possibility of reading information from heterogeneous controllers through BACnet and Modbus and combining it in a higher-level environment: "All data is available. We are using a server. A local server. […] So we combine all this data into Node-RED and make it [in a usable way]. So we could present it directly" (External Integration Vendor interview, personal communication, 2026). The site engineer agreed that bringing information together can make building management easier, but emphasized that data from different systems still has to be translated into a coherent structure.

This distinction was illustrated through the challenge of comparing similar variables across systems from different suppliers. A temperature value may be available from several ventilation

systems, but differences in naming, organization, units, context, or representation can make direct comparison difficult. The technical connection therefore solves only one part of the integration problem.

The site engineer's perspective suggests that the organization needs a common representation through which different sources can be understood, describing the harmonization problem in his own words:

> "The only problem I can see there is that Johnson Controls is giving the data in one type of format and another vendor is giving it in another type of format, and then it becomes a lot, a lot more harder to orientate yourself in the data. […] So the cognitive process I go through to get to that point is now changed, and I think that is like the main barrier of data-driven facilitation." (Site Engineer interview, personal communication, 2026)

Without such a representation, users may have to change the way they search for and interpret information depending on which subsystem they are looking at. The resulting workload is cognitive as well as technical.

The practical integration work reinforces this observation. The Airthings data could be retrieved through an API and represented within Node-RED, but further decisions were still required concerning which values should be displayed, how they should be organized, and what operational meaning should be attached to them. Similarly, the external vendor's ability to retrieve controller tags did not eliminate the need to understand what those tags represented in relation to the physical system.

The finding is therefore that technical interoperability is a necessary but incomplete condition for integrated building information. Integration also requires a form of harmonization that allows information from different sources to be interpreted consistently. This is particularly important where the intended users are not the specialists who originally configured the systems.

The distinction also affects the design of the proposed building-side architecture. A central processing layer can collect information from different sources, but the value of centralization depends on how the collected information is organized after it has been collected. The empirical material therefore points beyond the simple question of whether data can be integrated toward the question of what structure makes the integrated data understandable and useful.

6.7 Empirical synthesis

Across the six themes, a consistent pattern emerges. Majorstuhuset contains significant digital information, but this information is distributed across systems, platforms, documentation, and actor groups. Knowledge about the information is similarly distributed. As a result, people remain important in connecting technical systems with organizational action.

The empirical material suggests that fragmentation becomes most visible at the points where one actor or system must hand information to another. The handoff can involve locating data, identifying a responsible actor, explaining the situation, interpreting a technical value, or deciding what response is appropriate. These handoffs can be managed effectively when experienced personnel know the infrastructure, but they become more demanding when knowledge is missing, responsibility crosses boundaries, or systems use different representations.

Integration activities alter some of these conditions. API access can make data from an external service available within a building-side environment. BACnet and Modbus can provide technical access to legacy controllers. Node-RED can function as an intermediary environment through which information from different sources is processed. A building-owned edge layer can therefore reduce some of the separation between information sources without requiring immediate replacement of the underlying systems.

At the same time, the evidence shows that integration does not automatically remove the organizational problems associated with fragmentation. The building may still depend on external platforms, specialist knowledge, and human coordination. Technical accessibility does not automatically create semantic coherence. Visibility does not automatically create actionability. A common information layer can therefore change where coordination work occurs rather than eliminate coordination altogether.

The findings can be summarized through the empirical sequence shown in Table 4; Figure 2 in Chapter 7 maps this sequence onto the six themes described above.

Table 4. Empirical sequence linking fragmentation to operational action

Order	Condition
1	Distributed systems
2	Distributed knowledge

Order	Condition
3	Coordination requirements
4	Integration opportunities
5	Need for interpretation
6	Operational action

Source: Author's own elaboration based on the empirical findings in Chapter 6.

The sequence should not be interpreted as a causal conclusion at this stage. Rather, it summarizes the recurring relationships observed across the empirical material. The next chapter uses the theoretical framework developed earlier to interpret these relationships and examine how they answer the research questions.

In this way, the findings provide the empirical basis for moving from a description of fragmentation toward an explanation of governance capability. The central empirical observation is that the building does not suffer simply from too little digital information. The more significant issue is how information, knowledge, responsibility, and technical capability are distributed and connected within the building's existing digital environment.

7. Analysis

This chapter interprets the findings presented in Chapter 6 through the theoretical and conceptual framework developed in Chapters 2 and 3. The purpose is not to repeat the empirical material, but to use the theoretical concepts to explain what the observed patterns mean for the research questions. The analysis therefore moves between the case evidence and the concepts of digital infrastructure, layered modular architecture, ecosystem interdependence, platform governance, data governance, organizational coordination, and governance capability. Figure 2 summarizes the six empirical themes introduced in Chapter 6 and how they relate to governance capability before the interpretation below develops each connection in turn.

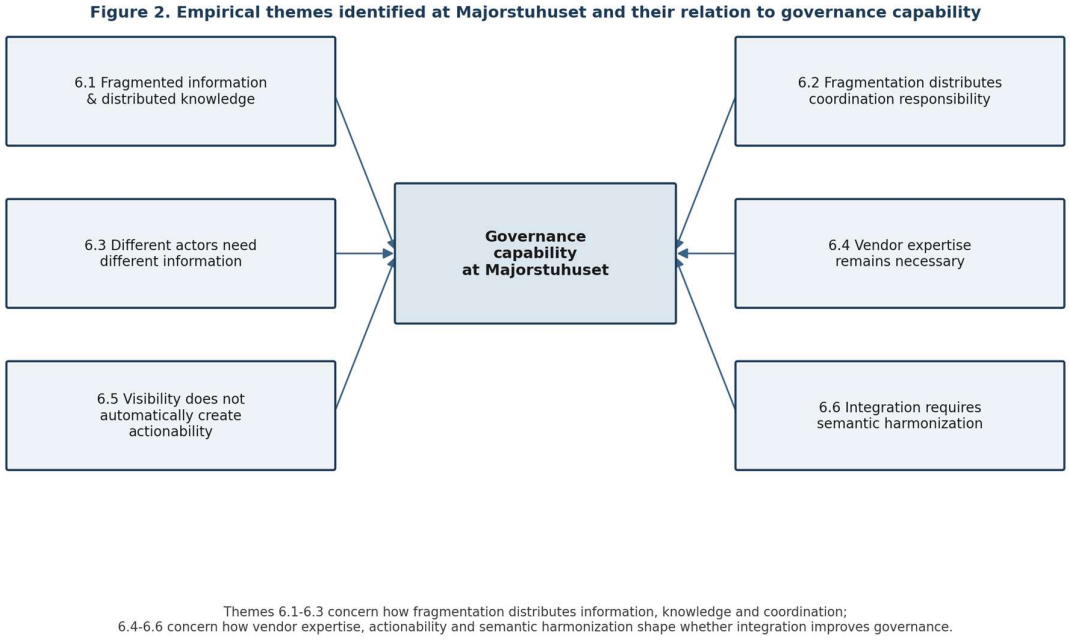


Figure 2. Empirical themes identified at Majorstuhuset and their relation to governance capability.

Source: Author's own elaboration based on the empirical findings in Chapter 6.

The analysis is deliberately separated from the empirical findings in accordance with the methodological design. No new empirical theme is introduced here; instead, the interviews, micro-ethnographic observations, system mapping, and technical investigation already presented in Chapter 6 are interpreted through the established framework. This distinction is important because the contribution of the analysis lies not in adding more data, but in explaining relationships between the patterns already identified.

7.1 Digital fragmentation as a governance condition

The findings suggest that digital fragmentation should be understood as a governance condition rather than merely as a technical inconvenience. Infrastructure theory provides the first explanation for this. Star and Ruhleder (1996) describe infrastructure as relational and embedded in practice, while Hanseth and Lyytinen (2010) emphasize that infrastructures evolve through installed bases, path dependency, and heterogeneous components. Tilson et al. (2010) further describe digital infrastructures as evolving arrangements that cross organizational boundaries. Taken together, these perspectives imply that fragmentation has consequences for how organizations access and coordinate information, not only for whether systems can technically communicate.

The Majorstuhuset findings support this interpretation. The operations manager described a building in which problems are frequently detected through manual inspection or tenant complaints and where the relevant service provider is identified through experience (Operations Manager interview, personal communication, September 7, 2026). The facilitator similarly described a situation in which numerous vendors each understand a particular part of the climate system, while a smaller internal group needs a guide capable of connecting this knowledge (Facilitator interview, personal communication, 2026). The engineer described technical documentation as fragmented and noted that project work can require substantial effort simply to reconstruct the state of existing systems (Site Engineer interview, personal communication, 2026). The pattern is therefore consistent across different organizational perspectives.

Fragmentation should consequently be understood as a condition in which the organization must repeatedly reconstruct a sufficiently coherent picture from partial views. A system can be functioning correctly within its own boundary and still contribute to a fragmented governance environment if the information it generates is difficult to connect with other information or with the actors responsible for decisions. This is consistent with the infrastructure perspective because infrastructures become problematic not only when they fail technically, but when the relationships through which they are used become difficult to maintain or extend (Star & Ruhleder, 1996).

The implication for the main research question is that fragmentation affects governance capability by increasing the distance between a building condition and an organizational response. The distance is not necessarily physical. It can be informational, interpretive, or organizational. Information may exist, but the relevant actor may not know where it is, how to

interpret it, or who should act on it. Governance capability is therefore shaped by the architecture of information as much as by the formal allocation of managerial authority.

7.2 Information distribution becomes knowledge distribution

One of the most important analytical developments from the case is that information fragmentation also becomes knowledge fragmentation. Infrastructure theory helps explain the first part of this relationship, but ecosystem and organizational perspectives clarify the second. When systems are developed and maintained by different actors, the contextual knowledge required to interpret them can remain attached to those actors. Jacobides et al. (2018) describe ecosystems as arrangements characterized by complementarities and multilateral dependencies. In Majorstuhuset, this means that different actors can contribute pieces of the knowledge required to operate the building even when no actor possesses the complete picture.

The site engineer provides the clearest example. The engineer explained that contractors often possess the most detailed knowledge of specific systems, while the building organization retains a broader overview. This creates a practical dependence: when the detailed information remains with the contractor, internal actors may lack sufficient grounds for an independent decision (Site Engineer interview, personal communication, 2026). The facilitator similarly described the need for trusted technical people who can bring knowledge from several vendors into the organization and help management judge whether a proposed solution is actually appropriate (Facilitator interview, personal communication, 2026).

This finding changes the interpretation of vendor dependency. The issue is not simply that a vendor controls a technical interface. The deeper issue is that system-specific knowledge can become attached to the supplier relationship. An open protocol may make a value technically readable, while the operational meaning of that value remains dependent on specialist knowledge. In other words, technical openness can reduce one type of dependency without removing knowledge dependency.

This also explains why a unified dashboard is not sufficient by itself. A dashboard can aggregate values while leaving their semantics distributed. The building owner may see a tag, but still not know precisely what equipment it belongs to, which physical area it affects, whether the reading is normal, or what intervention would be appropriate. The engineer explicitly described the need for information to be translated into understandable logic and for a semantic baseline that machines and humans can both use (Site Engineer interview, personal communication, 2026).

Documentation is therefore more than an administrative output. Within the theoretical framework, it can be understood as an infrastructural mechanism for retaining knowledge across organizational boundaries and over time. When documentation is missing, the organization must reconstruct the installed base through consultants, site visits, and vendor communication. The engineer described this reconstruction problem directly when asked what happens when a contractor with detailed system knowledge leaves the project:

> "They take their information with them. […] It has been times where the contractors have not done this. Then we are basically fumbling in the dark because we don't know what has changed, we don't know the status right now. We need an external consultant […] to map out the entire systems in a new way, and it creates a lot of headaches for us." (Site Engineer interview, personal communication, 2026)

The engineer's earlier description of having to spend two to three days categorizing existing building information for the digital-twin project provides a further empirical example of this reconstruction work (Site Engineer interview, personal communication, 2026). The governance implication is that documentation contributes to the organization's capability to remain independent enough to evaluate and coordinate external expertise.

This supports RQ1 by showing that infrastructural fragmentation distributes not only data but also interpretive capacity. The effect is therefore broader than information inconvenience. Fragmentation can shape who is capable of making sense of the building and, consequently, who is able to participate effectively in decisions concerning its infrastructure.

7.3 Coordination responsibility as an infrastructural consequence

The findings also indicate that fragmentation produces coordination work. The operations manager described current problem handling as heavily dependent on telephone and SMS communication, with experience being used to determine which supplier should be contacted. In some cases, a ventilation company must contact the heating or cooling supplier because responsibility falls between vendors (Operations Manager interview, personal communication, September 7, 2026). The site engineer likewise described projects that become significantly more difficult as more disciplines and actors become involved (Site Engineer interview, personal communication, 2026).

This can be interpreted through organizational coordination. Barley (1986) demonstrates that technologies can change interaction patterns and organizational roles, while Leonardi (2011) emphasizes how technologies and routines become intertwined over time. In Majorstuhuset, the absence of a coherent information environment means that people perform part of the

connective function between systems. Experience tells them whom to call; meetings establish shared understanding; vendors exchange information; and engineers reconstruct relationships between technical domains.

This human coordination can be described as a form of hidden infrastructure. It is hidden because it is not necessarily represented in a system diagram, yet the building depends on it to function. The operations manager's knowledge of which supplier handles which system performs a routing function. The engineer's ability to translate between technical disciplines performs an interpretive function. The facilitator's trusted technical relationships perform a governance function by helping management evaluate proposals. These human capabilities compensate for, but also reveal, gaps in the digital infrastructure.

The analysis should not interpret this coordination as inherently inefficient. The evidence suggests that experienced employees can make manual coordination work surprisingly well. The facilitator stated that the current system "works pretty well" even though physical inspections remain necessary (Facilitator interview, personal communication, 2026). Likewise, the operations manager explained that the lowest operational level often handles problems effectively because the janitor knows which company to contact (Operations Manager interview, personal communication, September 7, 2026). The issue is therefore not that human coordination should disappear. Rather, the question is where human effort is being used because it creates value and where it is being used because the infrastructure fails to provide a better information pathway.

This distinction helps answer the main research question more precisely. Digital fragmentation shapes governance capability when coordination becomes dependent on tacit knowledge, repeated handoffs, or manual reconstruction of responsibility. A governance-enhancing architecture should therefore aim to reduce unnecessary coordination while preserving the human coordination needed for ambiguous or complex cases.

7.4 Integration as infrastructural reconfiguration

The ThinkEdge intervention and the external vendor's integration approach can be interpreted as forms of infrastructural reconfiguration rather than replacement. This is consistent with Hanseth and Lyytinen's (2010) understanding of infrastructures as evolving installed bases and with Yoo et al.'s (2010) layered modular architecture. Instead of discarding existing systems, a new layer can be introduced to make selected information available across system boundaries.

The Airthings integration illustrates this logic particularly clearly. Sensor information generated in an external service could be accessed through an API, processed through the ThinkEdge and Node-RED, and presented in a new interface. The underlying sensing system remained in place. Likewise, the external integration vendor described reading data from legacy controllers through BACnet and Modbus and combining the resulting tags in a local server and Node-RED environment (External Integration Vendor interview, personal communication, 2026). In both cases, the integration layer sits above existing infrastructure.

Yoo et al. (2010) argue that layered modular architectures make digital components more loosely coupled and allow new services to emerge through recombination. This provides a theoretical explanation for why an intermediary layer can be attractive in an existing building. The building does not need to make every subsystem directly communicate with every other subsystem. Instead, selected information can be exposed to a shared service layer, reducing the need for point-to-point integration between all components.

However, infrastructural reconfiguration also creates a new component that has to be governed. A building-owned layer requires maintenance, security controls, documentation, access management, and resilience. The integration layer can reduce dependence on individual vendor interfaces while becoming a new dependency itself. This is consistent with the broader infrastructure literature: new infrastructure does not remove complexity; it reorganizes it (Hanseth & Lyytinen, 2010).

The analytical value of the ThinkEdge intervention is therefore not that it proves a particular architecture is optimal. Its value is that it demonstrates the possibility of adding a building-owned coordination layer while retaining the existing installed base. This creates an empirical bridge between the theoretical idea of layered modularity and the organizational problem of fragmented infrastructure.

7.5 Why integration does not automatically create value

The case also complicates any simple assumption that more integration produces more value. The theoretical framework treats data as an organizational resource whose usefulness depends on access, interpretation, and the ability to use it. Khatri and Brown (2010) argue that data governance involves decision rights and accountability around data, while the interoperability literature reviewed in Chapter 2 emphasizes that technical exchange does not guarantee semantic usability (Di Biccari et al., 2022; Qiang et al., 2023).

The empirical material strongly supports this distinction. The operations manager values information because it can allow intervention before damage or tenant dissatisfaction occurs. The facilitator wants enough information to know whether the building is functioning, but explicitly rejects unnecessary detail and warns against information overload. The site engineer wants richer data when it contributes to an engineering decision, but also emphasizes the need for semantic explanation and relevance to the task (Operations Manager interview, personal communication, September 7, 2026; Facilitator interview, personal communication, 2026; Site Engineer interview, personal communication, 2026).

This suggests that the use value of data is conditional rather than intrinsic. A temperature value has little governance value when it is isolated from location, time, equipment context, and responsibility. The same value becomes highly useful when it is linked to a ventilation condition, compared with historical behavior, or used to trigger a timely intervention. The value comes from the relationship among the data, the actor, the decision, and the action.

This is also where the distinction between use value and exchange value can be applied cautiously. The thesis does not attempt to use Marxist value theory as an empirical measurement framework. Instead, the concept of use value provides a useful conceptual reminder that the practical value of data is realized in use. The interviews indicate that management is willing to invest in digital systems when the information addresses a meaningful operational or economic problem, but is less interested in technical capabilities for their own sake. The facilitator repeatedly frames investments in terms of need and profitability (Facilitator interview, personal communication, 2026), while the engineer explains that the most useful information is information that allows concrete decisions about the building (Site Engineer interview, personal communication, 2026).

Integration should therefore be evaluated by the improvement it produces in the information-to-action relationship rather than by the number of systems connected. A smaller integrated information set that supports a critical operational decision may generate more organizational value than a much larger dataset that increases cognitive burden without improving action.

7.6 From visibility to actionability

The central conceptual chain of the thesis, set out in Figure 1 (Section 3.6), can now be interpreted against the case evidence: infrastructure configuration shapes information accessibility, which shapes interpretability, which shapes operational actionability, which in turn shapes governance capability.

The first transition concerns accessibility. If a system cannot expose information through a usable interface, that information cannot participate in building-level governance. The Airthings API and BACnet examples demonstrate mechanisms by which previously separate information can become accessible to a higher-level processing environment (External Integration Vendor interview, personal communication, 2026). The ThinkEdge investigation further demonstrates how building-side processing can create an intermediary point for selected information sources.

The second transition concerns interpretation. Accessibility alone is not enough. The site engineer explicitly described the need for data to be translated into understandable logic and for a semantic baseline that can be understood by humans and machines (Site Engineer interview, personal communication, 2026). This is consistent with Qiang et al. (2023), whose review demonstrates that different building ontologies and representations create challenges for interoperable analytics. Data can therefore be visible while remaining cognitively fragmented.

The third transition concerns actionability. The operations manager's emphasis on early warnings makes the temporal dimension clear: information must arrive early enough for intervention (Operations Manager interview, personal communication, September 7, 2026). The facilitator's preference for warnings about important failures rather than notifications about everything adds a relevance dimension (Facilitator interview, personal communication, 2026). Actionability consequently depends on timing and information selection as well as interpretation.

The final transition concerns governance capability. An actor may understand a condition but still be unable to act because authority, technical control, budget, or specialist support is absent. The external vendor made this condition explicit when describing his own integration philosophy: "If you can integrate it, but you can't control it, that's not good, you know. All my systems, I am controlling" (External Integration Vendor interview, personal communication, 2026). The engineer similarly described a situation in which detailed information was held by contractors and internal actors sometimes lacked sufficient grounds to make decisions (Site Engineer interview, personal communication, 2026).

The model therefore explains why fragmentation cannot be reduced to a connectivity problem. Different weaknesses can occur at different stages. A closed interface constrains accessibility. Inconsistent semantics constrain interpretation. Poor alarm logic constrains actionability. Lack

of responsibility or authority constrains governance capability. The model provides a layered explanation rather than a binary connected/disconnected classification.

7.7 Actor-specific information and layered governance

The findings further indicate that governance requires both integration and differentiation. The facilitator does not want every technical value, while the technical personnel need greater depth to communicate with vendors and diagnose systems (Facilitator interview, personal communication, 2026). The operations manager needs readable indicators and historical information, while the integration specialist may need detailed controller-level data. This suggests that a coherent information architecture should not be confused with one universal interface.

Yoo et al.'s (2010) layered architecture provides a useful way to interpret this. A common underlying information layer can support different services and presentations without requiring every user to interact with the same representation. Platform governance similarly suggests that participation can be structured through interfaces and access arrangements rather than by exposing the entire architecture equally to all actors (Tiwana, 2014).

This also connects to data governance. Khatri and Brown (2010) emphasize authority and accountability over data rather than unrestricted access. The empirical material supports a role-sensitive model: the owner or facilitator needs decision-relevant information; operations personnel need actionable operational information; engineers need technical and relational detail; and specialist vendors need deeper system information when performing maintenance or diagnostics.

Layered governance therefore combines a coherent internal information structure with differentiated external views. This can also support security because access can be organized according to legitimate operational roles. The goal is not maximum openness, but appropriate accessibility. This point is particularly consistent with the facilitator’s position that data sharing should depend on the importance of the data and that the system should remain secure without creating unnecessary restrictions on ordinary operational information (Facilitator interview, personal communication, 2026).

The analysis consequently challenges two extremes. A fully fragmented architecture creates unnecessary information barriers, while a fully flattened architecture can create information overload and inappropriate exposure. Governance capability requires an architecture that is

integrated enough to provide coherence but differentiated enough to match the needs and responsibilities of different actors.

7.8 Integration, authority and dependency

Integration changes dependency relationships but does not eliminate them. This is particularly visible when infrastructure is reconfigured through a building-owned integration layer. A local layer can reduce dependence on a vendor dashboard, but it creates a new dependency on the people who maintain that layer. Similarly, an open protocol can reduce interface lock-in while leaving the organization dependent on specialist knowledge of the equipment behind the interface.

The ecosystem perspective is useful here because it distinguishes dependence from productive interdependence. Adner (2017) and Jacobides et al. (2018) emphasize that innovation and value can depend on complementary actors rather than one organization controlling the complete system. The Majorstuhuset evidence supports this more nuanced interpretation. The facilitator explicitly recognizes the need for trustworthy vendors, while the engineer states that critical building operation remains strongly dependent on specialists: “I would say that we are totally dependent on them for the critical tasks. It’s some tasks you can’t get around, like the electrician” (Site Engineer interview, personal communication, 2026; see also Facilitator interview, personal communication, 2026).

The governance objective should therefore be **controlled dependency**. The building does not need to become completely independent from suppliers. Instead, it needs enough internal knowledge, documentation, architectural visibility, and decision rights to coordinate external expertise without losing strategic control. The engineer’s emphasis on maintaining the right information and the facilitator’s desire for a guide who can bring vendor knowledge into the organization point in this direction.

Control also has an authority dimension. The engineer argues that the building owner should retain ultimate control while trusted suppliers may retain operational responsibilities over their systems. The external vendor, in contrast, emphasizes technical control as necessary for meaningful integration. These perspectives suggest that governance depends on separating technical responsibility from ultimate decision authority. A supplier can operate a subsystem without becoming the final authority over the building’s digital infrastructure.

This distinction strengthens the connection between platform governance and building governance. The key question is not whether control is centralized or decentralized in absolute

terms. It is where different forms of control are located, who can override whom, and whether the building owner retains sufficient capability to understand and steer the system.

7.9 Conditions for governance-enhancing integration

The case supports six conditions under which integration is more likely to improve governance capability.

First, information must be accessible through stable and maintainable interfaces. APIs, BACnet, Modbus, and other mechanisms can provide technical access, but access must remain maintainable over time.

Second, information must be semantically harmonized. Comparable values should be identifiable and understandable across systems, with sufficient context about the equipment, location, unit, and meaning of each value.

Third, information must be timely enough for the decision it is intended to support. Early warnings can have much greater operational value than the same information received after a failure has already affected the building.

Fourth, information must be presented according to actor role and task. Strategic users may need summarized indicators, while engineers and vendors may require deeper technical detail.

Fifth, responsibility and authority must be connected to the information flow. A system should make clear who is expected to respond and whether that actor has sufficient authority and technical capability to intervene.

Sixth, the integrated architecture must remain secure, resilient, and governable. Greater connectivity should not create an unacceptable new operational vulnerability or simply replace one dependency with another.

These conditions synthesize the theoretical and empirical analysis. The first three primarily concern infrastructure and information; the fourth and fifth concern organizational governance; and the sixth addresses the need to balance accessibility with security and resilience. Together, they explain why integration is not an end state. Its value depends on whether it improves the conditions under which information can support organizational action.

Chapter 8 develops these relationships into a revised conceptual model (Figure 3) that also revisits the four analytical propositions introduced in Chapter 3, tying each proposition explicitly to a stage in the model and to the empirical evidence that supports it.

8. Discussion

This chapter discusses the findings in relation to the research questions and the theoretical framework. The purpose is not to repeat the empirical findings presented in Chapter 6 or the detailed theoretical interpretation developed in Chapter 7, but to establish what the case contributes to the broader discussion of digital infrastructure and governance in existing buildings. The discussion is organized around explicit answers to the main research question and the two sub-questions, followed by the theoretical contribution, the practical, methodological, ethical, societal and policy implications, and the boundary conditions of the findings.

8.1 Analytical propositions and a revised conceptual model

The theoretical framework developed in Chapter 3 leads to four analytical propositions, used here as sensitizing devices rather than as hypotheses to be statistically tested. Having presented the findings (Chapter 6) and the theory-led analysis (Chapter 7), this section revisits those propositions against the Majorstuhuset evidence and ties them explicitly to a revised conceptual model.

Proposition 1: Greater infrastructural fragmentation tends to distribute information and knowledge across systems and actors. This proposition follows from the installed-base and heterogeneous-infrastructure perspectives of Star and Ruhleder (1996), Hanseth and Lyytinen (2010), and Tilson et al. (2010).

Proposition 2: Distributed information increases coordination requirements where actors depend on one another to interpret or act upon information. This proposition connects infrastructure theory with organizational coordination (Barley, 1986; Leonardi, 2011).

Proposition 3: Infrastructural integration can reduce fragmentation-related coordination burdens when it creates meaningful and interpretable information flows. This proposition follows from the combination of layered digital architecture, infrastructure theory, and platform governance (Yoo et al., 2010; Tiwana, 2014).

Proposition 4: Operational actionability depends on more than visibility; it also requires timeliness, contextual interpretation, role relevance, clear responsibility, and sufficient ability to intervene (Khatri & Brown, 2010; Qiang et al., 2023; Di Biccari et al., 2022).

Figure 3 brings these propositions together with the conceptual chain first introduced in Figure 1, showing which proposition governs each transition from infrastructure configuration to governance capability.

Figure 3. Revised conceptual model of governance capability, with analytical propositions P1-P4

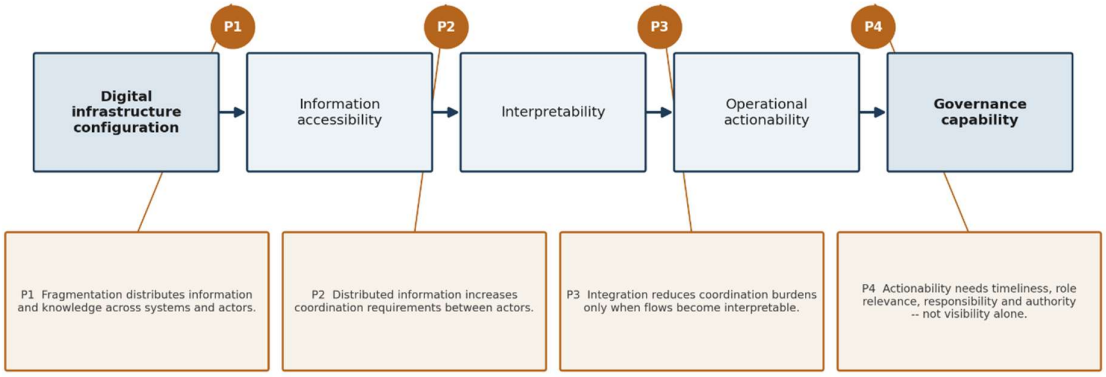


Figure 3. Revised conceptual model of governance capability, with analytical propositions P1-P4.

Source: Author's own elaboration, developed from Figure 1 and the empirical findings.

At the beginning of the chain, the installed and evolving infrastructure determines what information can be exposed and how easily new capabilities can be added (P1). Distributed access and knowledge then raise the coordination requirements examined under P2. Integration can strengthen the pathway from accessibility to interpretability, but, as P3 states, only when it also produces meaningful and interpretable information flows rather than simply adding connections. Finally, P4 makes clear that the transition from actionability to governance capability depends on timeliness, role relevance, responsibility, and authority, not on visibility alone.

Table 5 ties each proposition to the empirical evidence discussed in Chapters 6 and 7.

Table 5. Analytical propositions revisited against the Majorstuhuset evidence

Proposition	Empirical support at Majorstuhuset
P1 - Fragmentation distributes information and knowledge	Supported: information and interpretive knowledge were split between the facilitator, operations manager, vendor, and site engineer (Sections 6.1, 7.1-7.2).
P2 - Distributed information increases coordination requirements	Supported: manual detection and phone/SMS-based routing substituted for a shared information environment (Sections 6.2, 7.3).
P3 - Integration reduces coordination burdens only when flows are interpretable	Supported and refined: the Airthings and BACnet/Modbus work created accessibility, but semantic harmonization remained a separate, additional task (Sections 6.6, 7.4-7.5).
P4 - Actionability requires more than visibility	Supported: timeliness, role relevance, and authority to intervene each independently constrained actionability in the case (Sections 6.5, 7.6-7.7).

Source: Author's own elaboration based on Chapters 6 and 7.

Fragmentation acts primarily by disrupting the transitions shown in Figure 3. It can isolate information, attach knowledge to individual vendors, increase coordination work, and create uncertainty about responsibility. Integration can strengthen the pathway, but, consistent with P3 and P4, only if it also addresses semantics, role relevance, organizational authority, and long-term governance. The model is therefore intentionally more demanding than a simple interoperability framework: it moves the analysis from a technology-adoption question toward a capability-design question, and it does so with each of its claims traceable to a specific proposition and a specific body of empirical evidence.

8.2 Answer to the main research question

The main research question asks how digital fragmentation shapes governance capability in a multi-tenant commercial building. The evidence suggests that fragmentation shapes governance capability primarily by distributing information, knowledge, and coordination responsibilities across technical systems and organizational actors. The central problem is therefore not simply

that some systems are disconnected. It is that the organization must repeatedly bridge the resulting boundaries before information can contribute to a decision or intervention.

This finding extends the conventional interpretation of fragmentation. Building automation research commonly emphasizes interoperability and integration as technical challenges (Domingues et al., 2016; Verbeke et al., 2022). Infrastructure theory, however, suggests that heterogeneous systems should be understood as an evolving installed base embedded in organizational practice (Star & Ruhleder, 1996; Hanseth & Lyytinen, 2010). The Majorstuhuset case demonstrates the organizational consequence of this condition. The operations manager described manual detection and phone- or SMS-based coordination, while the site engineer described the effort required to reconstruct fragmented documentation and system knowledge.

Governance capability is consequently affected through the additional work required to assemble a sufficiently coherent picture of the building. Some of this work is visible, such as contacting vendors or commissioning a specialist. Other parts are less visible because they are embedded in experience: knowing which supplier is responsible, knowing which value is meaningful, or knowing when a deviation is significant. This supports Orlikowski's (1992) view that technology and organizational practice are mutually related and reinforces the relevance of organizational coordination to infrastructure studies.

The answer is therefore not that fragmentation makes governance impossible. Majorstuhuset continues to operate, and experienced personnel compensate for fragmented systems through routines, relationships, and specialist knowledge. Rather, fragmentation makes governance more dependent on these compensatory mechanisms. The greater the need to reconstruct information and responsibility through human coordination, the more governance capability depends on individuals and relationships that may be difficult to reproduce or transfer.

The main research question can therefore be answered directly: digital fragmentation shapes governance capability by increasing the distance between building conditions, usable information, organizational responsibility, and possible action. The effect is strongest where information is distributed, knowledge is concentrated in specific actors, and responsibility crosses organizational or system boundaries.

8.3 Answer to RQ1

RQ1 asks how fragmentation and integration affect the distribution of information, knowledge, and coordination responsibilities among actors. The case indicates that fragmentation distributes these dimensions unevenly. Information is located across multiple systems, while

detailed knowledge about those systems is often held by contractors, experienced personnel, engineers, or technical specialists. Coordination responsibility consequently falls on actors who have to connect these distributed resources.

The ecosystem perspective helps explain why this distribution persists. Adner (2017) and Jacobides et al. (2018) emphasize that ecosystems are characterized by interdependence among actors whose complementary contributions cannot always be brought under one hierarchy. Majorstuhuset reflects this pattern at the infrastructural level. The facilitator depends on technical personnel and trusted vendors; the operations manager depends on service providers for diagnosis; the engineer works across several specialist domains; and the integration vendor provides technical knowledge about legacy systems. The finding is therefore not that the building has failed to organize its actors, but that its organizational capability is distributed across an ecosystem of complementary expertise. The facilitator's observation that many vendors know their own areas but not necessarily the overall system illustrates this distribution particularly clearly.

Integration can alter this distribution by creating a shared information layer. Yoo et al. (2010) argue that layered modular architectures allow new digital services to be created without replacing every underlying component. The Majorstuhuset case supports this possibility through the ThinkEdge and Node-RED work, the Airthings API pathway, and the vendor's use of a local server to combine BACnet and Modbus information. Integration can therefore centralize access even where the underlying systems remain distributed.

However, the evidence shows that access centralization does not automatically produce knowledge centralization. The site engineer explicitly identifies a continuing problem in which contractors possess detailed system knowledge and the building organization may lack the information needed to make independent decisions. This is a significant qualification to the idea of integration. A common information point may reduce the number of interfaces that users need to access, while leaving the meaning of the underlying information distributed across vendors and technical specialists.

The answer to RQ1 is therefore twofold. Fragmentation distributes information physically and digitally, but it also distributes interpretive knowledge and coordination responsibilities. Integration can centralize access and reduce some handoffs, but it only changes the deeper distribution of responsibility when documentation, common semantics, and organizational roles are developed alongside the technical integration.

This also explains why the building's response should not be understood as a choice between complete internalization and complete outsourcing. The findings show value in external expertise, but also indicate the need for internal capability sufficient to coordinate and evaluate that expertise. This is consistent with Khatri and Brown's (2010) emphasis on decision rights and accountability and with ecosystem theory's distinction between useful interdependence and problematic dependence (Adner, 2017; Jacobides et al., 2018).

8.4 Answer to RQ2

RQ2 asks under what conditions integrated digital infrastructure can transform data visibility into operational actionability. The analysis identifies a sequence of conditions rather than a single technical requirement. Information must first be accessible; its meaning must then be understandable; it must arrive at a useful time; it must be relevant to the actor's role; and the actor must possess sufficient responsibility and ability to respond.

The first condition is technical accessibility. The Airthings integration demonstrates how an external information source can become available through an API, while the vendor's BACnet and Modbus work demonstrates how legacy controllers can expose information to an intermediary environment. These examples are consistent with the literature showing that interoperability can extend the possibilities of smart-building applications, but can also create new integration and governance challenges (Domingues et al., 2016; Verbeke et al., 2022).

The second condition is interpretability. This is where the Majorstuhuset case adds particular value. The site engineer does not describe the main obstacle simply as a lack of data. Instead, the concern is that equivalent information can be represented differently across systems, changing the cognitive process required to find and understand it. The engineer argues for semantic harmonization so that a sensor can be understood by both humans and machines. This finding is consistent with Qiang et al. (2023), who show that building ontology approaches differ and that semantic alignment remains necessary for data-driven applications.

The third condition is timeliness. The operations manager's emphasis on early warning shows that operational information has value only in relation to the time available for response. A cooling failure detected after tenant complaints is not equivalent to an alarm received early enough to intervene. The relevant criterion is therefore not merely whether the system can produce a value, but whether it can produce that value at the time the organization can still do something useful with it.

The fourth condition is role relevance. The facilitator explicitly rejects the idea that every actor needs the same amount of information and emphasizes a need-to-know approach. This supports the proposition that integration should not create one undifferentiated information space. Different actors require different abstractions of the same underlying information. Khatri and Brown's (2010) emphasis on decision rights is relevant here because appropriate access depends on organizational responsibility rather than on an assumption that maximum access is always beneficial.

The fifth condition is authority and capability to act. The external vendor explicitly argues that integrating a system without being able to control it is insufficient, while the site engineer describes situations in which internal actors cannot act because critical information remains with contractors. Visibility can therefore increase without increasing governance capability if the person receiving the information has no practical route to intervention.

The answer to RQ2 is therefore explicit: integrated digital infrastructure becomes operationally actionable when accessibility is combined with semantic interpretability, timeliness, role relevance, and sufficient responsibility and authority to respond. A unified dashboard is neither necessary nor sufficient in itself. What matters is whether integration reduces the practical distance between a relevant signal and an appropriate action.

8.5 Theoretical contribution

The theoretical contribution of the study is primarily integrative. Rather than proposing a new standalone theory of building digitalization, the thesis connects theoretical perspectives that are often discussed separately. Digital infrastructure theory explains why heterogeneous installed bases persist and why change is path dependent (Star & Ruhleder, 1996; Hanseth & Lyytinen, 2010; Tilson et al., 2010). Layered digital architecture explains how new services can be added without replacing every existing component (Yoo et al., 2010). Ecosystem and platform perspectives explain interdependence, modularity, interfaces, and architectural dependency (Adner, 2017; Jacobides et al., 2018; Tiwana, 2014). Data governance explains decision rights and accountability around information (Khatri & Brown, 2010).

The study connects these perspectives through the concept of governance capability. This concept is used here as an analytical synthesis: the ability of relevant actors to access, interpret, coordinate around, decide upon, and act on information necessary for managing the building. The contribution is therefore not the claim that these individual theoretical ideas are new, but

that their relationship provides a more precise explanation of why digital integration produces organizational value only under particular conditions.

The visibility–interpretability–actionability distinction sharpens this contribution. Existing interoperability discussions can establish whether information can move between systems, but the case shows that this does not establish whether actors can use the information effectively. The distinction therefore adds an organizational layer to the analysis of interoperability: connectivity concerns whether information can move; interpretability concerns whether it can be understood; actionability concerns whether it can support intervention.

This contributes to the literature by reframing digital building integration as a question of capability rather than connectivity alone. It also extends the use of platform and ecosystem concepts into an existing physical building context without treating the building itself as a commercial digital platform. The case demonstrates that architecture influences not only technical interoperability but also the distribution of knowledge, responsibility, and practical control.

8.6 Practical implications

For building owners and facilitators, the practical implication is to begin digitalization from operational decisions rather than from technologies. The first question should not be which systems can be connected, but which recurring problems or decisions are currently constrained by fragmented information. The operations manager's need for early warnings and the facilitator's need for concise indicators illustrate this decision-oriented starting point.

A second implication is that documentation should be treated as part of the digital infrastructure. The site engineer's difficulty locating and interpreting historical technical information shows that poor documentation creates direct organizational costs. Standardized documentation, naming conventions, system maps, and handover requirements can therefore increase governance capability even before advanced automation is introduced.

A third implication is that integration should be incremental and modular. A building does not need to replace every existing subsystem at once. A building-owned information layer can provide a controlled point for collecting selected information from heterogeneous sources. This follows the logic of layered modularity (Yoo et al., 2010) and is consistent with the practical experience in the case of using APIs and protocol-based integration.

A fourth implication concerns role-based information design. The facilitator's preference for useful, limited information and the engineer's requirement for deeper technical detail suggest that one interface should not be expected to serve all purposes. A more effective architecture would maintain a coherent underlying information model while presenting different levels of abstraction according to role.

A fifth implication is organizational capability. Integration should be accompanied by clear decisions concerning who maintains the integration layer, who owns documentation, who responds to alarms, and who retains authority to change or override systems. This is where data governance and platform governance become practical management concerns rather than purely technical matters (Khatri & Brown, 2010; Tiwana, 2014).

A useful implementation sequence emerging from the case is:

1. Identify high-value decisions and recurring failure situations.
2. Identify the minimum information required for those decisions.
3. Map where that information currently resides and who understands it.
4. Identify technical, semantic, and organizational barriers.
5. Introduce the smallest maintainable integration layer that addresses the priority problem.
6. Define information ownership, responsibilities, authority, security, documentation, and maintenance.
7. Evaluate the result according to improved actionability rather than increased data volume.

This approach can reduce the risk of developing a technically impressive system that creates more information without improving operational capability.

8.7 Methodological, ethical, societal and policy implications

Beyond the theoretical and practical implications above, the findings have methodological, ethical, societal and policy implications.

Methodologically, the study shows the value of combining interviews with direct observation and hands-on technical investigation. Several of the most important conditions in the case, such as the difference between information that is formally available and information that can be accessed in practice, became visible only when the researcher attempted to locate, connect or reorganize existing systems. This is consistent with the observation that infrastructure tends to become visible when someone tries to use or change it (Star & Ruhleder, 1996). Studies of building digitalization that rely only on interviews or documentation may therefore

overestimate how accessible and usable building information is. At the same time, the researcher's involvement in the technical work requires the kind of reflexivity and separation of evidence types described in Section 4.5.

Ethically, the case shows that building data is not neutral. Indoor-environment readings, occupancy-related signals, access-control records and camera systems can reveal information about the people who use a building, and integration brings such information together in fewer places. A building-owned integration layer therefore raises questions about the purpose for which data is collected, who may see which information, how long it is kept, and where it is stored. The facilitator's view that appropriate access depends on the sensitivity of the information, and the role-based information design proposed in Section 8.6, point toward an ethical principle of appropriate rather than maximal access. Integration also redistributes responsibility: when alarms and automated responses replace informal routines, it must remain clear who is accountable when a signal is missed or misinterpreted.

Societally, the ability of building actors to act on information affects energy use, indoor environmental quality and the reliability of the spaces in which people work. Data-driven building operation is widely seen as a route to better building performance (White et al., 2025), but the case suggests that this potential depends on organizational conditions as much as on technology. The concentration of system knowledge in individual suppliers and people also matters beyond a single building. Owners with limited internal technical capacity may find it difficult to evaluate proposals, change suppliers or retain knowledge over the long lifetime of building systems, which can slow the improvement of existing buildings more generally.

For policy, the findings suggest that requirements for building automation and data access address visibility more readily than interpretability and actionability. In the European Union, the recast Energy Performance of Buildings Directive extends requirements for building automation and control systems in larger non-residential buildings, establishes the smart readiness indicator as a common scheme, and requires that building owners, tenants and managers can access data from building systems and share it with third parties (European Parliament and Council of the European Union, 2024). Such measures can reduce technical barriers to access. The case indicates, however, that access alone does not secure governance capability. Public procurement guidance, building regulations and contractual practice could therefore also address documentation and handover requirements, consistent naming and information models, open interfaces, and clear allocation of data responsibilities, so that

building owners retain the ability to understand and act on the information their systems produce.

8.8 Boundary conditions

The findings should not be interpreted as evidence that centralization or maximum integration is always preferable. The facilitator's comments indicate that the appropriateness of data access depends partly on the sensitivity and importance of the information, while the engineer emphasizes the importance of keeping critical control sufficiently local and secure. Greater connectivity can create additional exposure, complexity, and responsibility, meaning that governance-enhancing integration must remain selective.

The same applies to vendor involvement. The case does not support eliminating specialist suppliers. The interviews consistently indicate that vendors provide valuable knowledge for complex technical tasks. The relevant boundary is between productive interdependence and excessive dependence. Building owners may benefit from specialist knowledge while still retaining sufficient architectural understanding, documentation, access, and decision authority to evaluate and coordinate that expertise.

Finally, the proposed model is most relevant to existing buildings with fragmented installed bases and distributed technical responsibility. A newly designed building with a coherent architecture from the outset may encounter different problems. The findings should therefore be transferred analytically, not mechanically. Their strongest relevance is to organizations facing the combination of heterogeneous legacy systems, multiple suppliers, limited internal technical capacity, and a need for incremental digitalization.

8.9 Discussion topics for future development

The discussion points above identify several directions for the final contribution of the thesis. First, the study contributes to the literature by connecting digital infrastructure, ecosystem interdependence, and governance through the information-to-action pathway. Second, it provides a practical framework for designing integration around operational decisions rather than around connectivity alone. Third, it highlights documentation and semantic harmonization as organizational infrastructure rather than secondary technical tasks. Finally, it shows that the governance value of digitalization depends on maintaining a balance between integration, differentiation, external expertise, security, and local control.

9. Conclusion

This chapter concludes the thesis. It summarizes how fragmented infrastructure shapes governance capability, what the study reveals about integration, and what this means for Majorstuhuset, before stating the contribution and limits of the study and its final conclusion.

9.1 From fragmented infrastructure to governance capability

This thesis examined how digital fragmentation of building infrastructure shapes governance capability in a multi-tenant commercial building. The Majorstuhuset case shows that the central difficulty is not simply that information is stored in different systems. The more consequential issue is that information, technical knowledge, responsibility, and authority can become distributed across systems and actors. The organization may therefore possess substantial amounts of data while still needing considerable human effort to determine what the data means, who should respond, and what action should follow.

The study consequently concludes that digital fragmentation should be understood as a socio-technical governance condition. It influences governance through the coordination work required to bridge system boundaries and through the dependencies created when knowledge about systems remains concentrated in particular people, vendors, or platforms. At the same time, the case demonstrates that fragmentation does not prevent effective operation. Experienced personnel, established supplier relationships, and specialist expertise can compensate for gaps in the infrastructure. The limitation is that such compensation can make governance dependent on tacit knowledge and relationships that are difficult to transfer, document, or scale. (Star & Ruhleder, 1996; Hanseth & Lyytinen, 2010; Tilson et al., 2010)

9.2 What the study reveals about integration

The study does not support the conclusion that more integration is automatically better. Instead, it shows that integration is valuable when it changes the conditions under which information can be used. The technical investigation demonstrated that information from existing systems can be brought into an additional processing layer without replacing the entire installed base. The Airthings API pathway and the legacy-system integration work illustrate different ways in which existing infrastructure can participate in a more coherent information environment. (Domingues et al., 2016; Verbeke et al., 2022)

The deeper conclusion, however, is that technical connectivity is only the beginning. A connected system can still leave users dependent on specialist knowledge, inconsistent terminology, or unclear responsibilities. The practical contribution of integration therefore lies

in reducing the distance between a relevant building condition and an appropriate organizational response. This is why the thesis distinguishes between connectivity, interpretability, and actionability rather than treating them as one outcome.

9.3 The governance capability created by the study

The study's central conceptual contribution is the formulation of governance capability as an information-to-action capacity, represented in Figure 3 (Section 8.1) and grounded in the four analytical propositions discussed there.

This formulation brings together the theoretical perspectives developed throughout the thesis. Infrastructure theory explains the installed and evolving technical environment; layered architecture explains how new capabilities can be introduced without replacing everything; ecosystem and platform perspectives explain interdependence and architectural dependency; and data governance explains the relationship between information, responsibility, and decision rights. The case demonstrates that these dimensions meet in practice at the point where an actor has to understand information and decide what to do with it. (Star & Ruhleder, 1996; Hanseth & Lyytinen, 2010; Yoo et al., 2010; Adner, 2017; Jacobides et al., 2018; Khatri & Brown, 2010)

The contribution is therefore not a claim that a new building-management technology has been invented. Rather, the thesis offers a way of evaluating digital infrastructure according to the organizational capability it creates. A system should not be considered successful simply because it exposes more data or connects more devices. Its significance depends on whether responsible actors can obtain relevant information, understand it in context, and act upon it with sufficient authority and competence. (Khatri & Brown, 2010; Tiwana, 2014)

9.4 Practical conclusion for Majorstuhuset

For Majorstuhuset, the most appropriate digitalization strategy is therefore incremental and decision-oriented. The case does not indicate a need to replace the existing infrastructure wholesale. Instead, improvement can begin with operational situations where fragmented information creates the greatest coordination burden. The building can then introduce targeted integration layers, common information structures, documentation, and role-appropriate interfaces around those needs.

The practical recommendation can be stated simply: start with the decision, not the technology. Identify the operational problem, identify the information required to address it, determine where that information currently resides, and then create the smallest maintainable integration pathway that closes the relevant gap. The architecture should be accompanied by clear

responsibilities for data, documentation, security, maintenance, and intervention. This allows the building to benefit from external specialist expertise without allowing critical knowledge and control to become unnecessarily concentrated outside the organization.

The study also points to documentation and semantic harmonization as strategic infrastructure. These activities may appear secondary compared with sensors, software, or dashboards, but they determine whether information remains understandable as systems, suppliers, and personnel change. For a long-lived building, preserving this knowledge can be as important as adding new digital functionality.

9.5 Contribution and limits

The thesis contributes an empirically grounded interpretation of digital fragmentation that connects building infrastructure with organizational governance. It shows how fragmented infrastructure can distribute information and knowledge, how this distribution creates coordination requirements, and how integration can improve governance capability only when technical and organizational conditions are addressed together. The findings extend beyond a simple technical interoperability perspective by emphasizing the pathway from information accessibility to operational action.

The conclusions should nevertheless be read within the boundaries of the study. Majorstuhuset is a single case, the interview sample is limited, and the broader integration architecture was only partially implemented. The study therefore does not establish universal effects or statistically measurable improvements in building performance. Its value is analytical: it identifies relationships and conditions that can be examined in comparable buildings with heterogeneous installed systems and distributed technical responsibilities.

9.6 Final conclusion

The most important conclusion of this thesis is that the governance value of digitalization lies not in making a building more connected, but in making it more understandable and actionable for the people responsible for it. Digital fragmentation becomes a governance problem when information and knowledge are difficult to connect with responsibility and action. Integration becomes meaningful when it closes that gap without creating disproportionate new risks or dependencies.

The Majorstuhuset case therefore suggests a shift in how existing buildings should think about digital transformation. The question is not simply, “How can all the systems be connected?” A more useful question is: “Which decisions are currently limited by fragmented information, and

what information architecture would allow the responsible actor to act more effectively?" This shift from connectivity to capability is the central conclusion of the study.

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Appendices

Appendix A — Interview Guide

The interview guide was organized around common themes while adapting emphasis to each participant's role. Themes included digital infrastructure, data accessibility, fragmentation, coordination, responsibility, vendor relationships, actionability, future digitalization, barriers and perceived value of integration. The guide explicitly asked participants whether bringing information together would improve actual decisions or actions rather than assuming that integration was inherently beneficial.

The main questions asked in each interview are listed below. They are taken from the interview transcripts and lightly edited for readability; follow-up probes and clarifications are omitted. Each interview opened with a short introduction to the study.

Building facilitator / board member

1. From your perspective as a facilitator and member of the board of directors, what are the most important things you need to know about the building in order to make a good decision?
2. A lot of data is generated in the building, for example by access control, CCTV, sensors and actuators. How do you see the value of data today?
3. When you have a dashboard with a lot of data, for example the Energy Control dashboard, it can be difficult to make sense of all the information. (Prompt for discussion.)
4. Imagine a dashboard showing, in very simple terms rather than graphs that require interpretation, whether the HVAC, fire alarm and access-control systems are working as they should. (Prompt for discussion.)
5. If information from these different systems could be brought together without replacing the existing systems, where do you think the greatest value would be?
6. How important is it where data is collected, where it is stored, and how it is used?
7. Can you explain your thought process when a vendor comes to you with a proposal?
8. If you could change one thing about how information and digital systems are managed in the building today, what would you change?
9. If you have twelve different vendors and something goes wrong, how do you know which vendor to call?

10. Is there anything about the value or use of building information that you think I have not considered, or anything else you would like to add?

Operations manager

1. Could you tell me about your role in the building and what your typical responsibilities look like? How long have you been involved with Majorstuhuset?
2. Are there things you keep track of on an everyday basis, for example regarding the ventilation systems?
3. Can you think of a recent situation where something was not working as expected and you had to find information or contact someone to understand what was happening?
4. If, for example, cooling or heating is not working as it is supposed to, how do you first notice the problem? Is it mainly that tenants complain, or is there another way?
5. Is there any information that is particularly useful for you, in case of a failure or during everyday operations? What would you like to see in a dashboard or application?
6. Can you explain how information flows between the different actors, for example from the janitor who inspects, to you, to the service provider or vendor?
7. How do you normally know which person or vendor to contact? Are there situations where it is not clear who is responsible for the job?
8. How much do you normally depend on external vendors to understand what is happening in the building?
9. When you receive information about something happening in the building, what makes that information useful to you? When a vendor fixes a problem, do you need detailed technical information or mainly a general overview?
10. How well do the systems and digital tools you currently use, such as the Energy Control dashboard and meters, support this kind of work? How often do you check the Energy Control dashboard?
11. If you had information from all the systems in the building available in one place, what would be most useful for you to see?
12. If you could change one thing about how information is currently handled in the building, what would you change, and why? Do you think there is value in bringing information together from different systems?
13. What happens to the knowledge when a company or a person is no longer involved, for example when a vendor is replaced?

14. Is there anything about how the building is managed, how information is handled, or the challenges you experience that I have not asked about and that you would like to add?

External integration vendor

1. Can you introduce your company and what you do in general?
2. Can you tell me about the project you are doing in the Majorstuhuset building?
3. When you started the work, what did you need to understand about the existing systems, and what has been the most difficult part of getting information from them?
4. What does BACnet allow you to do, and does it have limitations?
5. Once you connect to a system using Modbus or BACnet, what kind of data is available, and is it in a form that is useful for building management, or does it have to be converted?
6. Vendor lock-in: the Energy Control system has limited your ability to communicate with it. Is this a big problem when you deal with systems from other companies?
7. How important is documentation, for example of the existing controllers, when you connect to a system?
8. What are the criteria for making an integration project successful?
9. What makes some buildings more difficult to integrate than others?
10. Is actuation, being able to control the systems, also important in integration? What difference does it make?
11. What kind of controllers do you use, and do you have agreements with the system companies? Do you replace the existing controllers or add your own?
12. From your experience, what do people who do not work with these systems, for example building owners, usually not understand about integration?

Site engineer

1. Can you tell me about your role and the kinds of projects you have been involved in over the years?
2. How involved are you in choosing the technical solutions or systems that are used? Who is in the team, and how is the final solution decided?
3. How does a project start: who identifies the need, how do possible solutions appear, and what do you keep in mind when choosing between them? Do vendors usually offer one solution or several options?

4. How much do you think about the existing systems in the building when carrying out a project? Do you try to connect new systems to the old ones, or is it sometimes easier to keep them separate?
5. Can you describe the full workflow of a project of your choice: what you were trying to achieve, what was already there, what problems you ran into, and who had to be involved?
6. What information do you normally need about the building and its existing systems when you work on a project, and where do you get it? Do you sometimes have to ask external companies?
7. Who knows the building's technical systems best, and what happens when a person or company holding that knowledge leaves?
8. When choosing a new system or supplier, what are the most important things you look at? Do you consider its ability to communicate with existing and future infrastructure?
9. How much do you depend on suppliers when developing or changing systems, and what kind of knowledge do you get from them? Can the building work without them, how difficult is it to change supplier, and what would it take not to be dependent on them?
10. How important is it that the different technical systems communicate with each other? Can you give an example where this would be very useful, and are there situations where connecting systems is not worth the extra cost?
11. What information do you personally find most useful when working on a project, and what makes it useful? Have you ever had access to data but still needed someone to explain what it means?
12. Can you think of a situation where better information helped you make better decisions or solve problems?
13. If you could improve the digital systems in Majorstuhuset over the next few years, what would you change first?
14. Who should have control over the building's information and digital systems? Are there situations where suppliers should keep control over part of a system?
15. Looking at the building as a whole, is the biggest challenge with the different technical systems technical, organizational, financial or something else?
16. I have been working from the idea that bringing information from different building systems together could make the building easier to govern. Do you agree, do you see problems with this, or is there anything I might be missing?

17. Is there anything else you would like to add, or any advice you would like to give?

Appendix B — Interview Overview

Table B1 summarizes the four semi-structured interviews. All interviews were recorded, transcribed and then cleaned for readability (obvious recognition errors, repetitions and filler words removed without changing the meaning). Quotations in the thesis are taken from these cleaned transcripts.

Table B1. Overview of interviews

Participant (role)	Focus of the interview
Building facilitator / board member	Strategic governance, investment, ownership, responsibility and vendor relationships
Operations manager	Daily operations, failures, communication, information needs and vendor coordination
External integration vendor	Legacy systems, BACnet/Modbus, data access, local integration and technical dependency
Site engineer	Digitalization, documentation, semantic harmonization, contractor competence and future development

Source: Author's own elaboration.

Appendix C — Coding Framework

The analysis began with the initial codes listed in Section 4.9, which were then grouped into eight themes. Table C1 shows the grouping and where each theme is reported in the findings.

Table C1. Initial codes grouped into analytical themes

Theme	Description	Initial codes (Section 4.9)	Reported in
Fragmentation	Technical, informational, organizational and procedural fragmentation	infrastructure; fragmentation	6.1, 6.2
Accessibility	Where information resides and how actors obtain it	information access	6.1, 6.3
Interpretability	Semantic consistency, context and cognitive effort	information quality; interpretation	6.3, 6.5
Actionability	Timeliness, relevance, alarms, responsibility and ability to intervene	alarms; timeliness; actionability	6.5
Coordination	Handoffs, communication, vendor routing and interdisciplinary work	coordination	6.2
Dependency	Specialist knowledge, vendor relationships and documentation	vendor dependency; documentation	6.4
Governance	Ownership, decision rights, responsibility, security and strategic control	responsibility; control; security; decision-making	6.2, 6.4
Integration	Interfaces, protocols, centralization, harmonization and architectural reconfiguration	integration; semantic harmonization	6.6

Source: Author's own elaboration based on the coding of the empirical material.

Appendix D — Case Context: Infrastructure, Information Flows and Integration

This appendix gives the detailed description of the case that supports Chapter 5. It covers the building's existing digital infrastructure, the existing information flows, the ThinkEdge intervention, the Airthings integration, and the boundaries of legacy-system integration.

D.1 Existing digital infrastructure

The digital environment of Majorstuhuset contains several categories of systems rather than one integrated information platform. The empirical investigation encountered sensor platforms, ventilation-related systems, heating and cooling equipment, energy-related information, legacy controllers, cloud services, and digital interfaces. Some information is available through dedicated applications or APIs, while other information requires local access, specialist knowledge, manual inspection, or an intermediary integration process.

This heterogeneity is important because it means that the case cannot be accurately described using a simple connected-versus-disconnected distinction. A system can be digitally active and technically reachable while remaining organizationally separate. A sensor platform may expose information through a cloud API but still remain separate from the building's other operational systems. A legacy controller may communicate through BACnet or Modbus but still require

specialist interpretation. A building-management platform may provide a high-level view while information from certain underlying systems remains outside that view.

The project work therefore approached the infrastructure as a set of layers and boundaries. At the physical level, equipment and sensors collect or generate information. At the communication level, data may move through Ethernet/IP networks, industrial protocols, APIs, or other interfaces. At the processing level, systems such as Node-RED can reorganize information. At the application level, dashboards can present selected information to users. This layered view is consistent with the digital-architecture direction discussed in the research process and with the broader conceptualization of loosely coupled digital layers in Yoo et al. (2010).

The architecture being investigated was intentionally designed around the existing installed base rather than around complete replacement. A building-owned edge device could provide a point where information from selected systems is collected, processed, and presented. The potential benefit is that the building can gradually create new information pathways without requiring every existing subsystem to be replaced at the same time.

At the same time, the infrastructure contains boundaries that remain significant. Some systems are dependent on vendor platforms or specialist knowledge. Some data is stored outside the building. Some information is not exposed through standardized interfaces. The architecture is therefore better described as partially accessible and partially integrated than as fully centralized. This distinction is important for later analysis because it shows why governance capability cannot be inferred from the existence of digital systems alone.

D.2 Existing information flows

The case provides evidence of both digital and human information flows. Digital systems continuously produce information, but people remain important mediators between information and operational action. In the operational environment described by participants, a problem can be observed by a person, communicated to building staff, associated with a particular technical system, and then routed to a vendor. The resulting process can involve several communication and decision steps before the relevant intervention takes place.

The operations manager described a process in which incidents may be identified manually and communicated through telephone or SMS. Experience helps determine which service provider should be contacted. This means that knowledge about the building's vendor relationships functions as a practical routing mechanism. The person who knows which company is

responsible can reduce the time required to move from observation to action; when that knowledge is absent, additional coordination is required.

The process can be represented as the six steps shown in Table 3 (Section 6.2).

This sequence illustrates why information flow should not be considered only in terms of data transmission. The flow also contains organizational decisions. Someone must recognize that a deviation matters, determine which information is relevant, identify responsibility, and coordinate the response. Fragmented infrastructure therefore produces a hybrid information environment in which digital systems and human communication perform complementary roles.

The facilitator's perspective adds another layer. At a strategic level, there is interest in obtaining a building-level overview that makes important issues visible without requiring constant attention to multiple specialized systems. The objective is not to replace detailed technical information but to make the information necessary for decisions easier to obtain. This supports the broader distinction developed in Chapters 2 and 3 between visibility and actionability.

The case therefore contains a difference between information generation and information circulation. Systems can generate data automatically while the organization still relies on people to move meaning and responsibility across boundaries. The existence of an automated sensor does not imply automated governance. The organizational pathway surrounding the data remains important.

D.3 ThinkEdge intervention

The Lenovo ThinkEdge SE10 was introduced into the case as a building-side computing point intended to support a more coherent information architecture. The role of the device was not to replace tenant networks or the installed building systems. Instead, it was conceived as an additional layer through which selected information sources could be accessed, processed, and presented. This approach reflects the practical constraint of working with an installed base rather than designing the building's infrastructure from scratch.

The ThinkEdge was physically connected to the building network and configured for remote administration during the active phase of the project. Node-RED was installed as the environment for data processing and dashboard development. The broader architecture also envisioned MQTT as a communication mechanism between data-producing and data-consuming components. The purpose of this layered approach was to create greater flexibility:

data sources could remain independent while an intermediate layer provided common processing and presentation capabilities.

The technical intervention is analytically significant because it illustrates one possible form of infrastructural reconfiguration. Instead of replacing every existing system, a new building-owned layer can be placed between selected sources and applications. This corresponds with the layered modular architecture discussed by Yoo et al. (2010), where new digital services can emerge through recombination across relatively loosely coupled layers.

However, the ThinkEdge should not be described as a completed building-wide integration platform. The wider architecture remained staged and exploratory. Some elements were implemented, some were tested, and some remained proposed for future integration. The distinction is important because the thesis is concerned with the organizational implications of infrastructural integration rather than with presenting a prototype as a finished commercial solution.

The ThinkEdge experience also revealed the importance of access itself. Having a physical edge device does not automatically make every building system accessible. Each information source still depends on its own interface, credentials, protocol, cloud service, or physical connection. The edge device can therefore function as a coordination point without eliminating the underlying diversity of the installed infrastructure.

D.4 Airthings integration

Airthings provided one of the clearest examples of an existing digital information source that could be incorporated into the building-side architecture. The sensor platform produced environmental information including temperature, humidity, and carbon dioxide among other variables. Four accessible zones were used during the integration work, allowing the researcher to test how an existing cloud-connected information source could be brought into a local processing environment.

The integration relied on the Airthings cloud API rather than on a direct local sensor protocol. Authentication was established through the service's API mechanism, after which data could be retrieved and processed. Node-RED was then used to organize the retrieved values and present them through a dashboard. The practical information pathway is shown in Table D1.

Table D1. Airthings integration pathway

Step	Component
1	Airthings sensors
2	Airthings cloud
3	REST API
4	ThinkEdge
5	Node-RED
6	Dashboard

Source: Author's own elaboration based on the technical investigation.

This pathway demonstrates an important characteristic of infrastructural integration: the underlying sensor system did not need to be replaced for its information to participate in a new application. Instead, the integration layer interacted with an existing digital service. In this sense, the technical intervention was additive rather than substitutive.

The Airthings case also exposes a dependency that is easy to overlook in discussions of local edge computing. Although the ThinkEdge could process and display the information locally, the source remained dependent on the Airthings cloud service and its API. The integration therefore changed the location and organization of information processing without eliminating the external platform dependency.

The technical experience also reinforced the difference between data availability and governance value. Once the data was visible in Node-RED, further decisions were still required concerning what should be displayed, how it should be organized, what thresholds should be meaningful, and which user should receive the information. The technical connection was therefore a necessary step but not the final governance outcome.

The subsequent loss of remote access to the ThinkEdge also demonstrated another property of infrastructure research: practical access conditions can change during a project. The Airthings integration was later reconstructed in a separate environment for continued testing. This reconstruction allowed the information pathway and interface logic to be investigated further, but it is analytically distinct from the original building-side deployment and is therefore not treated as evidence of continuous operation at Majorstuhuset.

D.5 Legacy systems and integration boundaries

Legacy systems provide a contrasting case to the relatively accessible Airthings platform. The external integration specialist described work involving an older controller environment in which power was restored and communication with controllers was established through BACnet. Controller tags could then be retrieved and brought into a processing environment. This demonstrates that older equipment is not necessarily inaccessible simply because it predates newer digital services.

The case also indicates, however, that protocol accessibility is only one part of the integration problem. The external vendor's work depended on understanding the particular controller environment, its configuration, the available tags, and the relationship between technical values and the physical processes they represent. This makes specialist system knowledge an important part of the integration process even when an open protocol is available.

The integration work was associated with a local server and Node-RED environment through which information from systems associated with different vendors could be combined. This provides a second practical illustration of an intermediary architecture. Rather than requiring all underlying equipment to share a single platform, the intermediary can translate and organize information at a higher layer.

Energy Control represents another important boundary in the case. The presence of an existing building-system platform does not necessarily mean that every underlying system has become part of one coherent information space. Some older systems could be approached separately from the information already visible in Energy Control. The empirical case therefore distinguishes between a platform that provides an operational interface and a genuinely integrated information infrastructure spanning all relevant systems.

The site engineer adds the semantic dimension to this boundary. Even when information from different systems can be collected, vendor-specific representations can make the information difficult to compare or interpret. Similar measurements may have different naming conventions, structures, or contexts. This creates a second integration task after technical connectivity has been established: translating heterogeneous data into an operationally coherent representation.

These observations can be summarized as a sequence of increasingly demanding integration conditions, shown in Table D2.

Table D2. Increasingly demanding integration conditions

Order	Condition
1	Connectivity
2	Data access
3	Semantic interpretation
4	Operational relevance
5	Action

Source: Author's own elaboration based on interviews and the technical investigation.

The empirical case shows evidence at several points along this sequence. BACnet and Modbus can provide connectivity. APIs can provide data access. Node-RED can provide an environment for processing and representation. But the final transition toward operational usefulness depends on interpretation, context, responsibility, and action. This boundary between technical connectivity and semantic and organizational usability is central to the analysis that follows.

Appendix E — System Architecture and Implementation Status

The intended architectural direction is a building-owned logical layer above existing systems. The proposed design uses an edge-processing concept, a communication layer, protocol or API adapters and an application/dashboard layer. It is important to distinguish this architecture from the limited implementation completed during the project. Figure E1 shows the demonstrated data pathway and the elements that remained prepared or proposed, and Table E1 gives the status of each component.

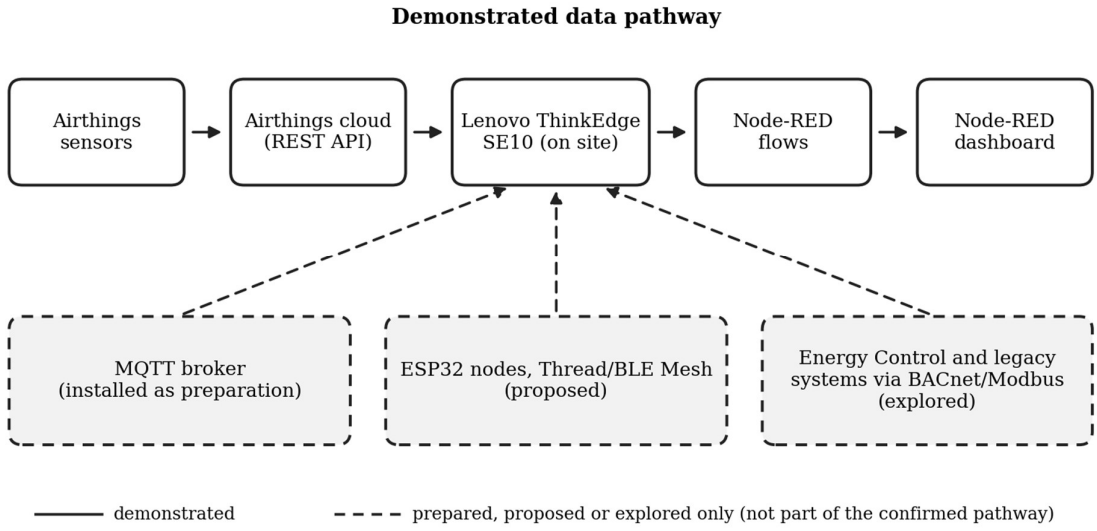


Figure E1. Demonstrated Airthings data pathway and prepared or proposed elements of the wider architecture.

Source: Author's own elaboration based on the technical investigation.

Table E1. Implementation status of architectural components

Component	Role in the architecture	Status
Airthings sensors and cloud service	Existing indoor-environment data source	Existing; accessed without modifying the sensor system
Airthings REST API	Data access from the cloud service	Demonstrated
Lenovo ThinkEdge SE10	Building-side edge-computing point	Installed at Majorstuhuset and connected to the building network
Node-RED flows and dashboard	Data processing and visualization	Implemented for the Airthings data
MQTT	Communication layer	Installed as preparation; not part of the confirmed Airthings data path
ESP32 nodes, Thread/BLE Mesh	Additional sensing and local communication	Proposed; not implemented
Energy Control and legacy systems (BACnet/Modbus)	Existing building systems and protocol access	Explored; integration work carried out by the external vendor, not by the researcher
Building-wide control functions	Automated control across systems	Not implemented

Source: Author's own elaboration based on the technical investigation.

Appendix F — AI Use

AI-assisted drafting was used during the preparation of this thesis for language organization, structural development, brainstorming and revision. Empirical claims remain the responsibility of the researcher and can be checked against original interviews, observations and source

documents. All technical implementation claims and participant statements can be verified against the underlying material before submission.